Abstract

Seedling detection is a crucial component of precision agriculture, enabling farmers to optimize crop management and resource allocation. Through the use of advanced imaging techniques such as computer vision and deep learning, seedlings can be accurately identified and monitored at early stages of growth. This technology allows for timely intervention in terms of irrigation, fertilization, and pest control, contributing to improved crop yield and resource efficiency. Our research will solve this problem of seedling detection and segmentation. Here we have shown the comparison between different YOLO models when our objective was only seedling detection. Later, we trained a segmentation model with our dataset. Although we got better results there, the model was re-trained with several modifications, in the model where we were able to increase the detection and segmentation accuracy a bit. The model achieved bounding box mAP of 0.682 and 0.906 at 50% IoU and segmentation mAP of 0.653 and 0.895 at 50% IoU.

Acknowledgements

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Chapter 1

Introduction

1.1 Problem Definition

Seedling detection plays a crucial role in plant science and agriculture.Early detection of seedlings allows to accurately estimate the number of plants that germinated and emerged successfully. This information is vital for understanding plant establishment, optimizing crop yield, and making informed decisions about resource allocation.By tracking the emergence and growth of seedlings over time, we can gain valuable insights into plant development, responses to environmental stimuli, and potential effects of stress factors. This information helps us improve breeding practices, develop better crop management strategies, and predict future yields.

Manual seedling detection, while a traditional and widely used method, is very labourous and time-consuming process and have many drawbacks and challenges. Counting seedlings by hand can be a slow and tedious process, especially for large fields or dense plantings. This can be physically demanding and impact the efficiency and productivity of farm work.

Manual counting of seedling can be subjective and prone to human error, such as missing seedlings, overcounting due to fatigue, or misidentifying weeds as seedlings. This can lead to inaccurate estimates of plant populations and impact decisions based on that data.

Figure 1.1: Manually detecting and counting seedling in the field [1]

Depending on a number of variables, including the size of the area, the density of seedlings, and the environmental conditions, counting seedlings by hand can be time-consuming and labor-intensive. The following elements affect how challenging it is to manually count seedlings:

- Large area: If the area where the seedlings are planted is vast, it will take a considerable amount of time to cover the entire area and count each seedling.
- High density of seedlings: When seedlings are densely planted, it can be challenging to count them accurately. Seedlings can be easily miscounted or overlooked due to their close proximity to one another.
- Visibility: If the seedlings are small, covered by other vegetation, or planted in an area with poor visibility, it can be difficult to spot and count them.
- Uneven terrain: Counting seedlings in an area with uneven terrain, rocks, or other obstacles can be physically demanding and increase the risk of miscounting.
- Weather conditions: Adverse weather conditions, such as extreme heat, cold, or rain, can make manual seedling counting even more challenging and uncomfortable.

In recent years, advances in technology, such as drones and computer vision algorithms, can lead to the development of automated methods for plant seedling detection, which can save time and improve accuracy.

Weeds are hazardous for cultivation, posing a significant threat to their productivity and livelihoods. Weeds compete with crops for sunlight, water, and nutrients, significantly impacting growth and ultimately reducing yields.

Manual weed detection and classification in agriculture relies on labor-intensive visual surveys, where farmers or workers inspect fields to identify and remove weeds. However, this approach has significant drawbacks. It is time-consuming, requiring substantial human resources and increasing the risk of errors. Subjectivity and variability in weed identification pose challenges, especially for non-experts, leading to inconsistent weed management practices. The method's limited scalability results in missed infestations, impacting crop health in large agricultural areas. Inefficient early weed detection delays appropriate measures, causing competition for resources and reduced crop yield. Dependence on human expertise introduces variability and limitations, and the labor-intensive nature leads to increased costs for farmers, impacting overall profitability. To overcome these challenges, the adoption of advanced technologies, such as remote sensing, computer vision, and machine learning, is crucial. Automated weed detection systems can reduce labor requirements, increase accuracy, and enable early detection, thereby enhancing weed management practices and optimizing agricultural productivity.

Figure 1.2: Manual Weed Detection [2]

Object detection in complex environments such as agricultural fields presents a lot of challenges. These challenges arise due to diverse factors including variable lighting conditions, overlapping plants, and the non-uniform appearance of weeds and seedlings. The emergence of deep learning has brought a new era of solutions to these challenges, with convolutional neural networks (CNNs) [22] leading the forefront in image recognition tasks. Recent literature, including notable studies like DeepSeedling [7] has demonstrated the efficacy of deep learning in similar applications. Nevertheless, there remains an expansive scope for exploration and enhancement in this sector. Moreover, studies like DeepWeeds [6] uses deep learning model to classify different kinds of weeds show potential application of deep learning in this sector.

This paper delves into the exploration of various deep learning architectures aimed at seedling detection in cotton fields. We benchmark our methodologies against existing standards and shed light on our findings and innovations. Through our research, we aspire to find the way for efficient and more accurate solutions for cotton farmers worldwide, emphasizing the broader goal of maximizing crop yield using cutting-edge technology.

We successfully compiled a comprehensive dataset of cotton seedlings, sourced directly from the Bangabandhu Sheikh Mujibur Rahman Agricultural University (BSMRAU). Leveraging the cotton seedling dataset, we trained three prominent YOLO [23] object detection models: YOLOv5, YOLOv8, and YOLONAS. Our experiments provided valuable insights into the capabilities and performance nuances of each model when applied to agricultural datasets. For a more granular analysis, we introduced Cascade Mask R-CNN. This model was trained to perform instance segmentation, providing precise portrayal and labeling of individual instances within the agricultural images.

The primary contributions of this study include:

1.Dataset Collection of Seedling:We have compiled a dataset of cotton seedlings from the Bangabandhu Sheikh Mujibur Rahman Agricultural University (BSMRAU). This dataset contains 714 high quality images of cotton field at seedling stage. Images are collected by Umanned Ariel Vehicle(UAV).

2.Weeds Classification: We developed a model which can classify weeds into seven different types of weed class. The model is developed on DeepWeeds [6] dataset which can classify a weeds.

3.Seedling Detection and Segmentation: After compiling the cotton seedling dataset, we trained it with different object detection and segmentation models, along with architectural modifications to perform detection and instance segmentation of the seedlings.

1.2 Motivation Towards Our Work

In the 21st century, agriculture remains critically important as it tackles various challenges and shapes our future. It serves as a key driver of economic growth, especially in developing nations, providing livelihoods for millions and fostering rural development. Efficient farming practices not only stimulate economic diversification but also create new opportunities. Agriculture plays a crucial role in managing ecosystems, preserving biodiversity, and mitigating climate change. Sustainable practices contribute to soil conservation, water pollution reduction, and carbon sequestration, promoting a healthier planet. Interconnected with global challenges like poverty, hunger, climate change, and water scarcity, agriculture's innovative solutions can address broader issues, paving the way for a more resilient future. Its significance lies in ensuring food security, driving economic growth, protecting the environment, and overcoming global challenges. Embracing sustainable practices, technological advancements, and collaborative efforts can lead to a future where agriculture thrives, benefiting both people and the planet.

The world's food production needs to be increased by 70% by 2050 (Fig. 1.3) compared to 2010 level to feed the growing population of the world, which will become roughly 9.2 billion [24] [25]. But the arable land will increase by only 5% [3] (Fig. 1.4). Moreover, food production will face many challenges in the coming decades, like scarcity of water [26], climate change [27] etc. Also, food consumption per person will be increased in this timespan [5] (Fig. 1.5). In this circumstance, maximizing the yield of the crop is the only way to overcome this challenge. Food production has to be increased without negatively impacting the environment. For example, carbon emissions cannot be increased, soil quality cannot be reduced, water supplies cannot be depleted, and the use of pesticides has to be limited.

In traditional farming practices, resources such as fertilizer, pesticides, water, and nutrients are uniformly distributed across the entire field, resulting in inefficiencies and unnecessary costs. This approach fails to consider the variability of specific areas within the field that may have distinct requirements. However, in the face of the current challenges posed by climate change, precision agriculture emerges as a promising solution. By adopting precision agriculture techniques, farmers can precisely tailor the application of resources to specific areas based on their unique needs, optimizing resource utilization and minimizing waste. Precision agriculture represents a pivotal component of the ongoing evolution in modern agricultural practices, often referred to as the third wave of agricultural revolutions. Its implementation holds significant potential for enhancing productivity, sustainability, and the overall resilience of our agricultural systems.

In the agricultural sector, the use of artificial intelligence can add a new dimension. To meet the challenges of agriculture in the 21st century, the use of modern technology like deep learning and machine learning is inevitable.

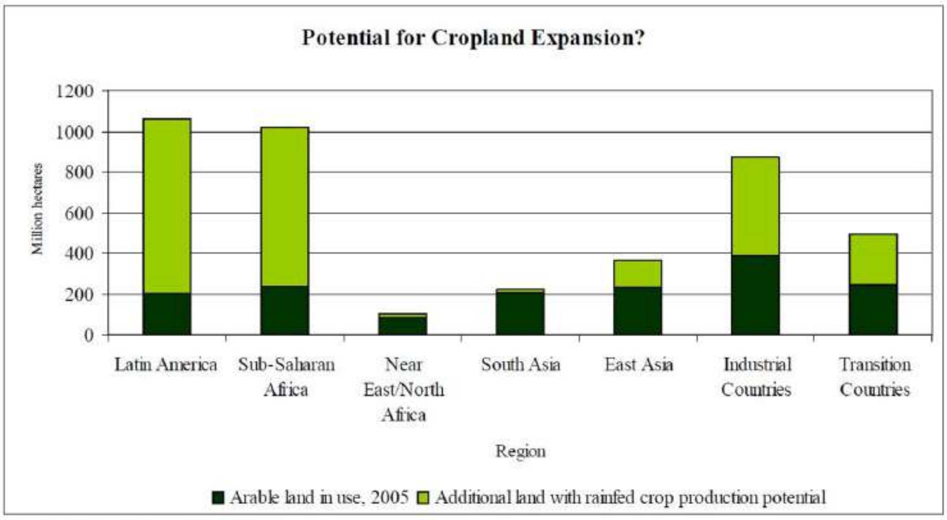
Modern technology like deep learning and machine learning can overcome challenges in modern agriculture and optimize crop management practices. Deep learning, a subset of artificial intelligence, offers powerful capabilities in image analysis and pattern recognition, making it a promising tool for addressing these challenges.

Maximizing crop yield entails addressing two significant challenges: weeds and plant population density. These factors pose significant obstacles to cultivating healthy and vibrant crops, directly impacting overall yield. Improper plant density and weed infestations are detrimental to the agricultural sector, as they substantially reduce productivity. Conventional cultivation methods often result in the inefficient use of resources, time, and labor, leading to suboptimal land utilization. Furthermore, these conventional practices are unsustainable and environmentally unfriendly. Therefore, it is imperative to develop effective solutions that can tackle weed control and optimize plant density in a more efficient, sustainable, and environmentally conscious manner.

Weed infestations pose a significant threat to crop health and productivity. Manual weed detection and control methods are often labor-intensive, time-consuming, and subjective. Deep learning provides an opportunity to automate the process by training models on extensive datasets of weed images. These models can learn to accurately identify and classify different weed species, enabling targeted and efficient weed management strategies. By detecting weeds at an early stage, farmers can intervene promptly, reducing competition for resources and minimizing the adverse effects on crop yield.

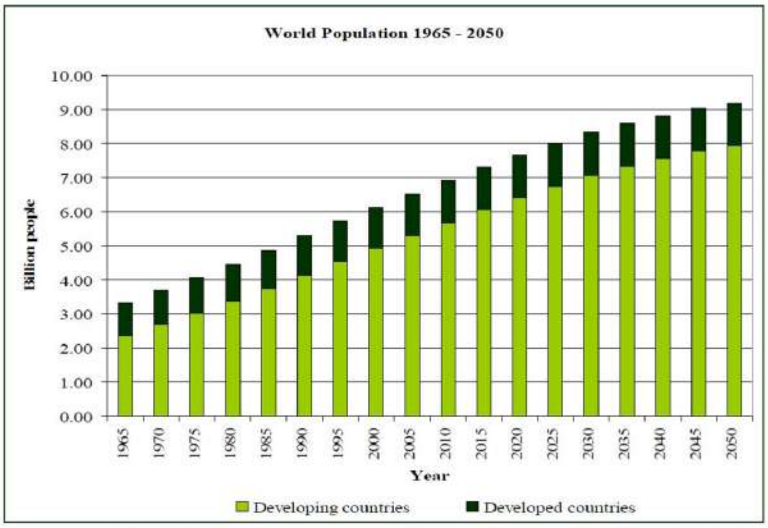
Seedling counting is crucial for optimizing planting strategies and assessing crop health. Traditional manual counting methods are prone to errors and can be impractical for large-scale agricultural operations. Deep learning algorithms, combined with image processing techniques, offer the potential for automated and precise seedling counting. By accurately determining seedling populations, farmers can make informed decisions regarding irrigation, fertilization, and plant spacing, leading to optimized plant density and improved resource utilization.

The motivation for employing deep learning in weed detection, seedling counting which can facilitate measuring plant seedling density is driven by the desire to enhance agricultural productivity, reduce costs, and promote sustainable farming practices. By harnessing the power of deep learning, farmers can effectively combat weed infestations, optimize plant density, and improve overall crop management, leading to increased yields, reduced resource wastage, and more sustainable agricultural systems. These advancements have the potential to revolutionize agricultural practices by improving weed management, enhancing planting strategies, and achieving efficient resource utilization. By leveraging deep learning techniques, farmers can make informed decisions, optimize crop yield, and contribute to sustainable agricultural development, ultimately leading to increased productivity, reduced costs, and a more environmentally friendly approach to farming.



Source: Bruinsma (2009).

Figure 1.4: Possible Expansion of Cropland [3]



Source: Population Division of the Department of Economic and Social Affairs of the United Nations Secretariat (2007)

Figure 1.3: World Population by 2050 [4]

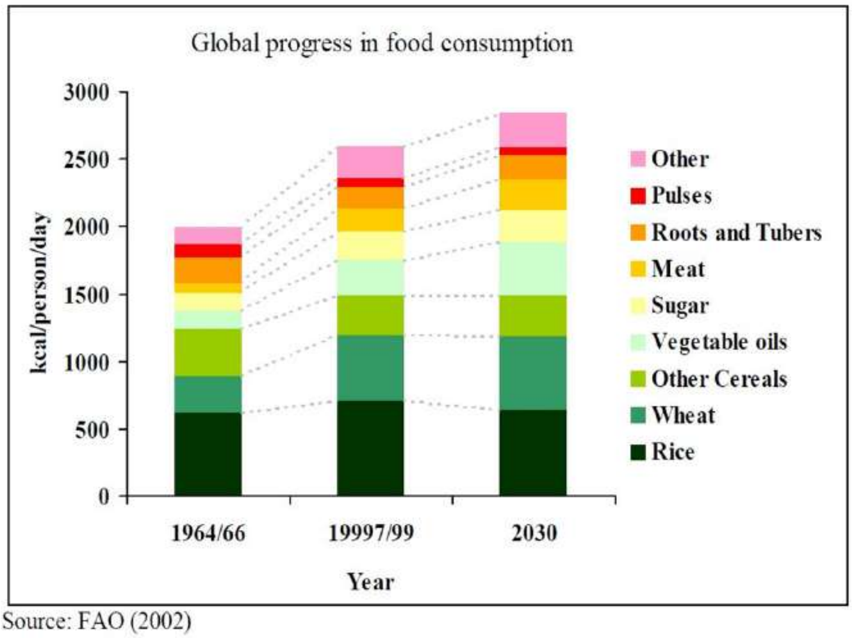


Figure 1.5: Graph of Food Consumption by Years [5]

1.3 Objective

To address the challenges faced by the agriculture sector, we have developed a framework that instills new optimism for the industry. This innovative framework centers around optimizing plant population density and implementing early-stage weed eradication measures. By precisely managing plant density and proactively identifying and eliminating weeds in their early stages, our framework aims to enhance agricultural productivity and sustainability. Through the application of advanced techniques and strategic interventions, we aspire to empower the agricultural sector with effective solutions that contribute to its growth and resilience.

Accurately estimating plant population density can be facilitated through the process of seedling counting. By counting seedlings at an early stage of plant growth, the estimation of plant population density becomes more feasible and reliable. Seedling detection serves as the foundational step in this approach, enabling the subsequent task of seedling counting. This systematic approach provides valuable insights into plant density, allowing for informed decision-making and effective management practices in agriculture.

Weeds pose a significant challenge to attaining optimal crop yield as they directly impact crop productivity. As yield-reducing agents, weeds have adverse effects on both the agricultural and economic aspects of crop production. Timely and accurate detection of weeds plays a crucial role in safeguarding the agriculture sector from this detrimental problem. By identifying and addressing weeds at an early stage, proactive measures can be implemented to mitigate their negative impact, ensuring the overall health and productivity of crops.

The objectives of this project are:

- To develop a precise deep learning-based technique for seedling detection
- Classify different types of weeds
- Proper use of agricultural land through plant density optimization based on the seedling detection and count
- Using deep learning to maximize the crop yield
- Eco-friendly and sustainable development of agriculture

The proper implementation of robust procedures for plant density optimization and early weed detection in modern and precision agriculture is instrumental in cultivating a favorable perception of these agricultural practices. By employing advanced techniques and technologies, such as optimizing plant density and employing early weed detection strategies, farmers can enhance crop productivity, reduce resource wastage, and promote sustainable farming methods. This holistic approach not only improves overall agricultural efficiency but also contributes to environmental stewardship and long-term agricultural sustainability.

Chapter 2

Related Works

2.1 DeepWeeds: A Multiclass Weed Species Image Dataset for Deep Learning

2.1.1 Description

In this paper [6] a multiclass weed species image dataset is introduced specifically designed for training and evaluating deep learning models in the domain of weed detection and classification.

DeepWeeds [6] consists of a diverse collection of images encompassing ten different weed species commonly found in agricultural fields. The dataset includes variations in lighting conditions, growth stages, occlusions, and overlapping instances to mimic real-world scenarios encountered in farming environments. A total of 18,000 high-resolution RGB images are meticulously annotated at the pixel level to provide precise weed species segmentation masks.

To demonstrate the utility of DeepWeeds, multiple state-of-the-art deep learning models are trained and evaluated, including convolutional neural networks (CNNs) [28] and semantic segmentation networks. The performance of these models is assessed based on metrics such as accuracy, precision, recall, and F1-score. The effectiveness of different data augmentation techniques compare to enhance the model's generalization capabilities.

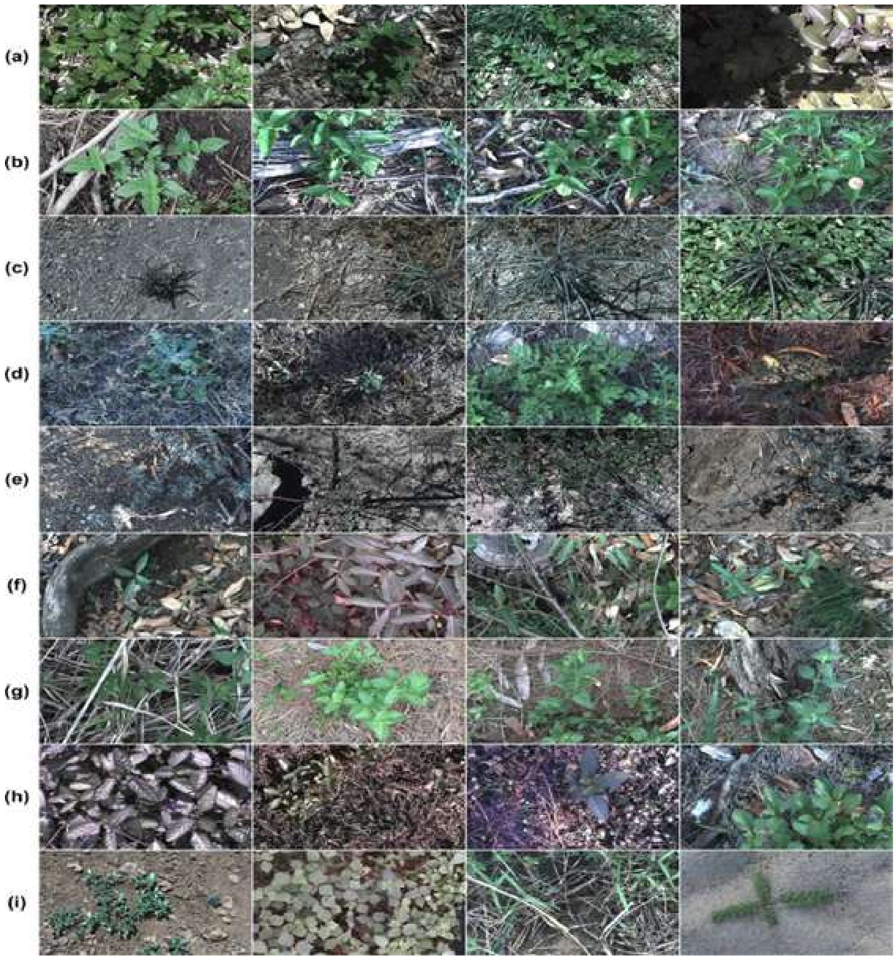


Figure 2.1: Sample images from each class of the DeepWeeds dataset [6]

The results show that deep learning models trained on the DeepWeeds dataset achieve remarkable performance in weed species detection and segmentation tasks. The highest-performing models consistently achieve accuracies above 90% across all weed classes. Moreover, the incorporation of data augmentation techniques further improves model robustness, reducing the risk of false positives and false negatives in practical applications.

To encourage collaboration and foster advancements in the field of weed detection, the DeepWeeds dataset is publicly available to the research community. Researchers and practitioners can leverage this dataset to develop and evaluate novel deep learning models, benchmark their performance, and explore innovative solutions for weed management in agriculture.

2.1.2 Strength

- The paper introduces the DeepWeeds dataset, which is a large-scale, multi-class dataset comprising images of various weed species.This dataset consists of 17509 high-resolution RGB images of different kind of weeds. The dataset covers a wide range of weed species commonly encountered in agricultural fields, making it a valuable resource for training and evaluating weed detection and classification models.

- The DeepWeeds dataset is annotated with precise bounding box annotations for each weed instance, providing ground truth information for training deep learning models. The accurate annotations enable the development of robust weed detection and segmentation algorithms, ensuring the reliability and quality of the dataset.

- The paper addresses the challenge of multiclass weed species classification, which is crucial for targeted weed management. By including a diverse range of weed species in the dataset, the paper facilitates the development and evaluation of deep learning models capable of accurately classifying multiple weed species simultaneously.

- The highest-performing models consistently achieve accuracies above 90% across all weed classes. Moreover, the incorporation of data augmentation techniques further improves model robustness, reducing the risk of false positives and false negatives in practical applications.

  Accuracy(Inceptionv3 [29]) = 95.1%

  Accuracy(ResNet50 [30])=95.7%

  Precision=95.7%

2.1.3 Weakness

- Limited diversity in environmental conditions: The DeepWeeds dataset has limitations in representing a wide range of environmental conditions that exist in different agricultural regions. Variations in lighting conditions, soil types, and cultivation practices can impact the appearance and characteristics of weeds. Therefore, a more diverse representation of environmental conditions would enhance the dataset's applicability across different agricultural settings.

- Imbalance in class distribution: Class imbalance, where certain weed species have significantly more or fewer instances compared to others, is a common challenge in multiclass datasets. The paper does not explicitly address the issue of class imbalance in the DeepWeeds dataset. This imbalance can affect the performance of machine learning models, leading to biases and reduced accuracy for underrepresented classes.

- Limited weed growth stages: The DeepWeeds dataset focuses on images of mature weed plants, potentially overlooking variations in weed appearance at different growth stages. Weed detection and management often require monitoring and identification at early growth stages, where weeds might have distinct characteristics. Including images capturing different growth stages would enhance the dataset's usefulness in comprehensive weed management.

- Lack of detailed metadata: The paper does not provide extensive metadata associated with the images in the DeepWeeds dataset. Additional information, such as geolocation, date of capture, and specific cropping systems, would enable researchers to analyze the impact of location, time, and cropping practices on weed growth and distribution, thus providing more context for analysis and interpretation.

- Lack of performance evaluation: The paper focuses primarily on the creation and description of the DeepWeeds dataset, but does not provide a comprehensive evaluation of the performance of deep learning models trained on the dataset.

2.2 DeepSeedling: Deep Convolutional Network and Kalman Filter for Plant Seedling Detection And Counting in the Field

2.2.1 Description

Accurate and efficient plant seedling detection and counting are critical tasks in various agricultural applications, including crop monitoring, yield estimation, and weed management. In this paper [7], deepseedling, a novel framework is proposed that combines deep convolutional networks (CNNs) [28] with a Kalman filter for robust and real-time plant seedling detection and counting in field environments has been proposed.

The DeepSeedling framework begins with the training of a CNN using a large-scale dataset of annotated plant seedling images. The CNN is designed to learn discriminative features and spatial relationships that enable accurate seedling detection even in challenging conditions, such as occlusions, varying lighting conditions, and complex backgrounds. By leveraging the power of deep learning, the CNN achieves high precision and recall rates, ensuring reliable seedling identification.

To further enhance the robustness and accuracy of the seedling counting process, a Kalman filter is incorporated into the DeepSeedling framework. The Kalman filter effectively tracks the detected seedlings over time, leveraging temporal information to improve counting accuracy and handle potential occlusions and movement. This fusion of deep learning and filtering techniques provides a comprehensive solution for seedling detection and counting in dynamic field environments.

Here, the performance of DeepSeedling is evaluated on a diverse set of real-world datasets captured in different agricultural settings. The results demonstrate the effectiveness of our framework in accurately detecting and counting plant seedlings across various crop species. Compared to existing methods, DeepSeedling achieves superior performance in terms of both detection accuracy and counting precision, while maintaining real-time processing speeds suitable for on-site applications.

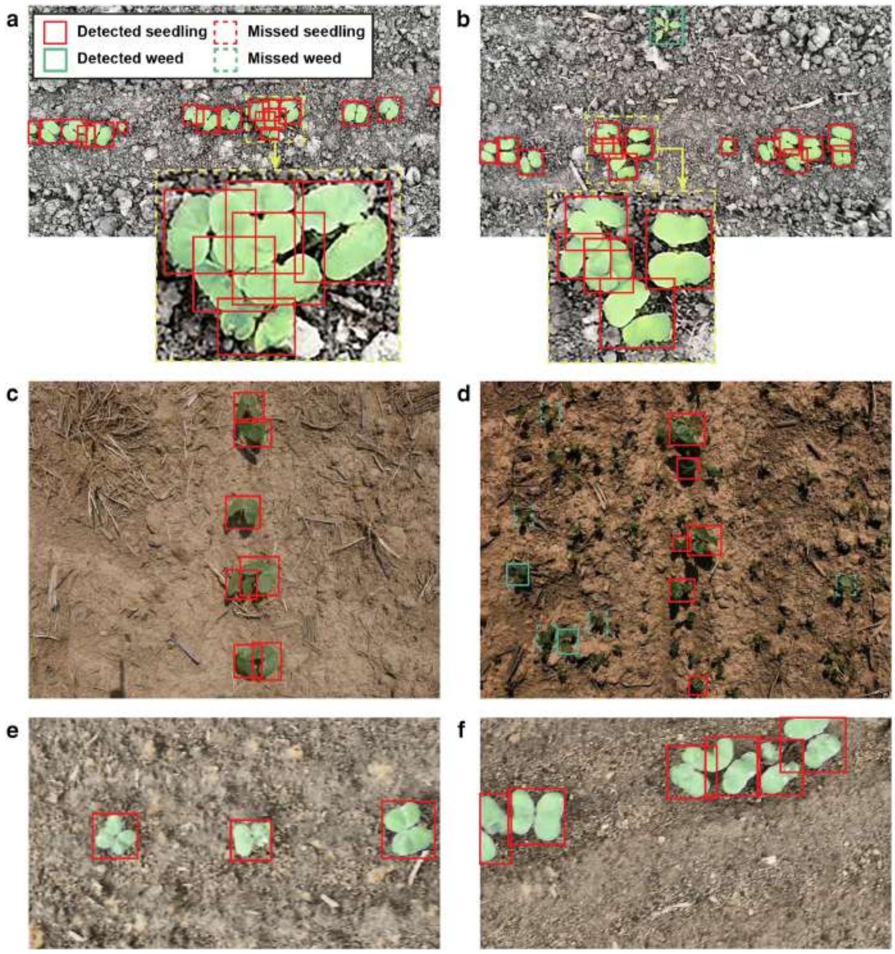


Figure 2.2: Seedling detection on field [7]

Additionally, extensive experiments is conducted to analyze the robustness of DeepSeedling under challenging conditions, including varying lighting conditions, occlusions, and varying plant growth stages. The results showcase the resilience of the proposed framework and its ability to adapt to different environmental factors, making it suitable for practical deployment in the field.

2.2.2 Strength

- Integration of deep learning and Kalman filter: The paper proposes a hybrid approach that combines deep convolutional neural networks (CNNs) with a Kalman filter for plant seedling detection and counting. This integration allows for the benefits of both techniques, leveraging the discriminative power of deep learning for accurate detection and the temporal filtering capabilities of the Kalman filter for improved tracking and counting.

- Development of a specialized dataset: The authors created a dataset specifically tailored for plant seedling detection and counting in the field. This dataset includes annotated images of various plant species at different growth stages, providing a reliable ground truth for training and evaluating the proposed system. The specialized dataset ensures the applicability and relevance of the developed approach to the target task.

- Accurate detection and counting performance: The paper demonstrates the effectiveness of the DeepSeedling system through comprehensive experiments and evaluation. The proposed approach achieves high accuracy in plant seedling detection and counting tasks, outperforming traditional methods and showcasing the capabilities of deep learning and the Kalman filter in addressing the challenges associated with counting plant seedlings in complex field conditions.

- Robustness to occlusions and variations: The DeepSeedling system shows robustness in handling occlusions, overlapping seedlings, and variations in lighting and background conditions. This robustness ensures reliable detection and counting even in challenging scenarios, where traditional methods may struggle to accurately identify individual seedlings.

- Real-time processing capability: The proposed system is designed to operate in real-time, allowing for efficient and timely counting of plant seedlings in the field. The combination of deep learning and the Kalman filter enables fast and accurate processing, facilitating prompt decision-making and effective management strategies for farmers and researchers.

2.2.3 Weakness

- Limited generalizability: The paper does not explicitly discuss the generalizability of the DeepSeedling system to different plant species or field conditions. The performance and effectiveness of the proposed approach may vary depending on the specific characteristics of the plant species and the environmental conditions in which the seedlings are being detected and counted.

Further exploration and evaluation across a broader range of plant species and field conditions would strengthen the applicability of the system.

- Dependency on labeled data: Deep learning models typically require large amounts of labeled data for training. The paper does not address the challenges associated with acquiring and labeling sufficient training data for different plant species and growth stages. The availability of labeled data may be limited, time-consuming, and resource-intensive, potentially hindering the adoption and scalability of the DeepSeedling system.

- Limited analysis of computational requirements: The paper lacks a detailed analysis of the computational requirements of the DeepSeedling system. Deep learning models can be computationally demanding, and the integration with the Kalman filter adds additional computational complexity. The paper does not provide insights into the computational efficiency or hardware requirements, which may impact the system's real-time processing capabilities and practical feasibility.

- Lack of comparison with alternative methods: The paper does not provide a comprehensive comparison of the DeepSeedling system with alternative methods for plant seedling detection and counting. A comparative analysis against existing approaches, including traditional computer vision methods or other deep learning-based techniques, would provide a better understanding of the strengths and weaknesses of the proposed system and its advantages over alternative solutions.

- Lack of real-world deployment validation: While the paper presents promising results through experiments and evaluation, it does not include a validation of the DeepSeedling system's performance in real-world field conditions. The system's accuracy and robustness in real-world scenarios, which may involve complex field conditions, diverse plant species, and varying growth stages, would provide stronger evidence of its practical applicability and reliability.

2.3 Weed and Corn Seedling Detection in Field Based on Multi Feature Fusion and Support Vector Machine

2.3.1 Description

This research paper[8] proposes a method for automatically identifying and detecting weeds and corn seedlings in fields using a combination of multi-feature fusion and Support Vector Machines (SVM) [31]. The goal is to enable precision farming techniques such as targeted herbicide spraying and fertilization by accurately locating both crops and unwanted weeds.

Weeds compete with crops for resources and can significantly reduce yield. Traditional weed control methods often involve blanket herbicide application, which harms the environment and wastes resources. Accurately identifying and locating weeds allows for targeted herbicide spraying, reducing environmental impact and improving efficiency.

In this paper, a technique is used which is called k-means clustering in lab color space to break down the field image into different regions. This helps separate individual plants from the background soil and other objects.The system then examines each of these regions to gather key information about the plants within them. This includes:

- Color features: Looking at the distribution of colors to identify different types of plants
- Texture features: Analyzing how the intensity of colors varies across the surface of the plants, which can help distinguish between smooth leaves and rough bark, for example
- Shape features: Measuring the size and shape of the plants, as different plant species often have distinct forms

The system merges these different types of information together to create a more complete picture of each plant. This is like putting together different pieces of a

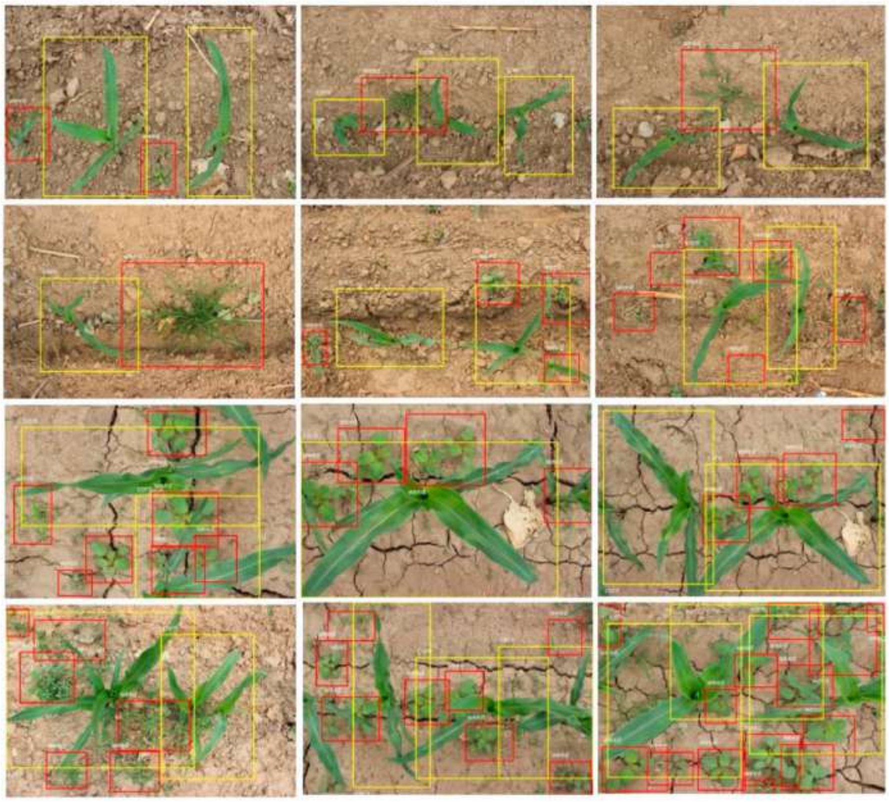


Figure 2.3: Seedling and weed detection on field [8]

puzzle to reveal a clearer image. A trained machine learning algorithm called a Support Vector Machine (SVM) then analyzes this combined information to decide whether each region contains a corn seedling, a weed, or just background. Once the system has identified the weeds and corn seedlings, it pinpoints their exact locations within the field. This allows farmers to target their weed control efforts precisely, reducing the need for herbicides and protecting the environment.

2.3.2 Strength

- Accuracy: The proposed method achieves a high accuracy in differentiating between corn seedlings, weeds, and background, exceeding previous approaches based on Gabor wavelets and Gradient Field Distribution. The reported accuracy of 97% is promising for practical applications.
- Robustness: The multi-feature fusion approach appears to be robust to variations in lighting conditions, which is a significant challenge in field environments. This is essential for real-world deployment of the system.

- Flexibility: The system allows for flexibility in terms of feature selection and fusion strategies. This enables further optimization and adaptation to different types of crops and weeds in diverse agricultural settings.
- Computational efficiency: The authors claim that the proposed method is computationally efficient, making it suitable for real-time implementation in agricultural robots and precision farming systems.
- Novelty: While other studies have used SVM for weed and crop detection, the combination of multi-feature fusion with SVM in this specific context appears to be novel and effective.
- Potential for practical applications: The paper demonstrates the potential of the proposed method for practical applications in precision farming, including targeted herbicide spraying and fertilization. This could lead to significant benefits in terms of environmental sustainability and agricultural efficiency.

2.3.3 Weakness

- Limited data: The study used a relatively small dataset for testing the method. Although the reported accuracy is encouraging, further validation with larger and more diverse datasets is crucial to confirm the generalizability and robustness of the system in real-world scenarios.
- Crop and weed specificity: The method's effectiveness may vary depending on the specific types of crops and weeds present in different fields. More research is needed to assess its performance across a wider range of agricultural settings with diverse plant species.
- Computational complexity: While the authors claim efficiency, implementing the system in real-time on agricultural robots or machinery might require further optimization and adaptation to specific hardware and software limitations.
- Environmental factors: The paper primarily focuses on static images under controlled conditions. The system's accuracy and reliability could be affected

by dynamic factors like lighting changes, shadows, and wind-induced plant movement. Additional testing under various environmental conditions is necessary.

- Generalizability of feature fusion: The paper compares different fusion strategies to find the most effective one for a specific dataset. However, the optimal method might not be universal and require adaptation for different crops and scenarios

2.4 Rice Seedling Detection in UAV Images Using Transfer Learning and Machine Learning

2.4.1 Description

The paper [9] uses a technique for automating rice seedling detection in agricultural fields using Unmanned Aerial Vehicles (UAVs) equipped with cameras. This technology holds the promise of revolutionizing rice farming by enabling:

Identifying and locating rice seedlings allows for targeted interventions like weed control and fertilization, minimizing resource waste and optimizing crop yields. UAV-based detection significantly reduces the time and labor required for traditional manual seedling counting, freeing up resources for other crucial tasks. The collected data can be used to generate maps of seedling distribution, providing valuable insights for informed decision-making throughout the farming cycle. Rice seedlings are often tiny and visually similar to weeds and background elements, making manual detection laborious and prone to errors.

UAVs capture high-resolution images of rice fields from above. Transfer learning: Pre-trained deep learning models like EfficientDet and Faster R-CNN are fine-tuned to identify rice seedlings specifically. The models analyze various features within the images, including color, texture, and shape, to differentiate seedlings from other elements. Based on the extracted features, the models classify each pixel in the image as either rice seedling, weed, or background. Identified rice seedlings are precisely located within the field, providing their GPS coordinates.

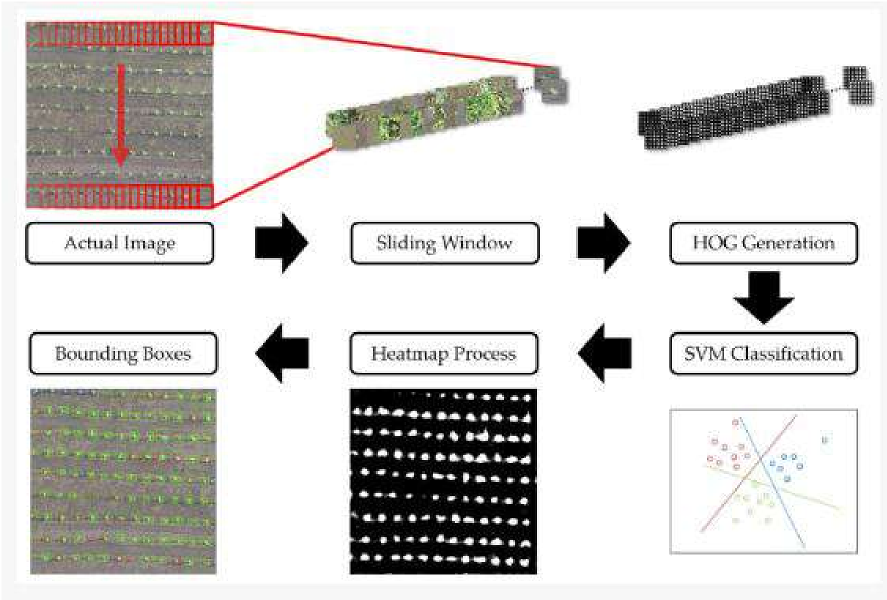


Figure 2.4: A diagram of rice seedling detection based on HOG and SVM [9]

The paper reports promising results, with the models achieving over 95% accuracy in detecting rice seedlings. This suggests the potential for reliable and effective implementation in real-world agricultural settings. The authors highlight areas for further research, such as expanding to diverse rice varieties and environmental conditions, optimizing models for real-time processing on resource-constrained UAVs, developing integrated systems for automated weed control and fertilization based on seedling detection data.

Overall, this paper presents a significant step forward in automating rice seedling detection, paving the way for more efficient, data-driven, and sustainable rice farming practices.

2.4.2 Strength

- Accuracy: The proposed method achieves high accuracy in differentiating rice seedlings from weeds and background, exceeding previous approaches based on simpler features. Reported accuracy above 95% is promising for practical applications.
- Robustness: The transfer learning approach using pre-trained models like EfficientDet and Faster R-CNN [32] appears to be robust to variations in

lighting conditions, a significant challenge in field environments. This is essential for real-world deployment.

- Efficiency: The paper claims computational efficiency, making the method suitable for real-time implementation on UAVs. This could enable real-time weed control and data collection during field surveys.

- Novelty: While other studies have used deep learning for weed and crop detection, the combination of transfer learning with UAV images specifically for rice seedlings appears to be novel and effective.

- Data-driven decision making: The detailed location data of rice seedlings opens up possibilities for generating field maps and analyzing seedling distribution, supporting data-driven agricultural management decisions.

- Potential for practical applications: The paper demonstrates the potential of the proposed method for practical applications in precision agriculture, including targeted herbicide spraying and fertilization. This could translate to significant benefits in terms of resource efficiency and environmental sustainability.

- Addressing a crucial challenge: Rice seedling detection is a vital aspect of rice farming, and automating this process can significantly improve farm efficiency and crop yields.

- Openness to future improvements: The authors acknowledge limitations and highlight areas for further research, such as adapting the model to different rice varieties and environmental conditions. This shows a commitment to continuous improvement and refinement of the technology.

Overall, the paper presents a promising new approach for rice seedling detection with several strengths and potential for real-world applications. While further research is needed to address some limitations, the potential impact on the future of rice farming is undeniable.

2.4.3 Weakness

- Limited dataset: The study used a relatively small dataset for training and testing the models. Although the reported accuracy is encouraging, further validation with larger and more diverse datasets, including different rice varieties, weed species, and environmental conditions, is crucial to confirm generalizability and robustness in real-world scenarios.
- Specific rice varieties: The effectiveness of the models might not be equally strong across all rice varieties, particularly rare or local cultivars with unique characteristics. More research is needed to adapt the models to diverse rice morphologies and growing conditions.
- Environmental variations: The paper primarily focuses on static images under controlled conditions. The system's accuracy and reliability could be affected by dynamic factors like lighting changes, shadows, wind-induced plant movement, and different soil textures. Testing under various environmental conditions and adapting the models to handle them is necessary.
- Computation on UAVs: While claimed to be efficient, implementing the models on resource-constrained UAVs in real-time might require further optimization, especially for larger fields or higher resolution images. Balancing accuracy with processing speed for practical UAV deployment is important.

2.5 Digital Counts of Maize Plants by Unmanned Aerial Vehicles (UAVs)

2.5.1 Description

Accurate and efficient plant counting plays a crucial role in various agricultural applications, including crop monitoring, yield estimation, and precision farming. Traditional manual counting methods are time-consuming, labor-intensive, and prone to human error. In this paper [10], a novel approach is proposed for digital

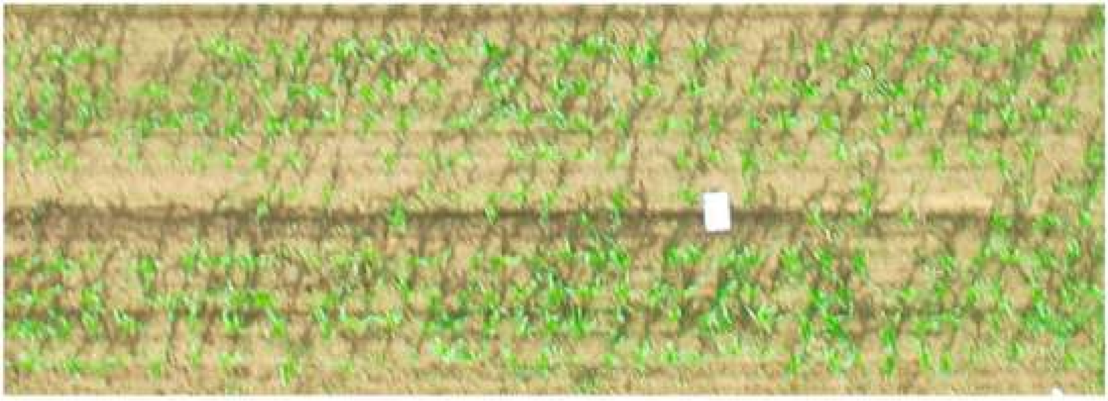


Figure 2.5: Drone view of a maize field [10]

counting of maize plants using Unmanned Aerial Vehicles (UAVs) equipped with imaging sensors.

The method leverages the high-resolution aerial imagery captured by UAVs to detect and count individual maize plants in large-scale fields. A robust image processing pipeline is developed that integrates computer vision algorithms and machine learning techniques to automate the counting process. The pipeline consists of multiple stages, including image preprocessing, plant detection, and plant counting.

In the image pre-processing stage, some techniques are applied such as image rectification, normalization, and enhancement to optimize the quality and consistency of the aerial imagery. This preprocessing step ensures accurate plant detection and counting in diverse environmental conditions and lighting variations.

For plant detection, advanced computer vision algorithms are employed, including object recognition and segmentation, to identify individual maize plants within the aerial images. A deep convolutional neural network (CNN) is trained on a large annotated dataset to learn discriminative features and achieve high accuracy in plant detection. The CNN effectively handles variations in plant appearance, scale, and orientation, making it robust to different field conditions.

Once the plants are detected, efficient counting algorithms are utilized to determine the number of maize plants in the field. Techniques such as clustering, region-based counting, and density estimation are employed to accurately estimate plant counts while accounting for overlapping and occluded plants.

The performance of this method is evaluated on multiple UAV-captured datasets of maize fields. The results demonstrate that this approach achieves high accuracy and efficiency in plant counting compared to manual counting methods. The digital counts obtained from this method exhibit strong correlation with ground truth counts, indicating its reliability and accuracy.

2.5.2 Strength

- Use of unmanned aerial vehicles (UAVs): The paper leverages the use of UAVs for capturing aerial imagery of maize fields. UAVs provide a unique and advantageous perspective, enabling efficient data collection over large areas in a relatively short time. The utilization of UAVs in the study showcases the potential of aerial imagery for crop monitoring and management.

- High-resolution aerial imagery: The paper emphasizes the acquisition of high-resolution aerial imagery, which allows for detailed and accurate analysis of maize plants. High-resolution imagery provides finer-grained information, enabling precise identification and counting of individual plants. This aspect enhances the accuracy and reliability of the digital counting approach proposed in the paper.

- Automated plant counting algorithm: The paper presents an automated plant counting algorithm specifically designed for maize plants. The algorithm takes advantage of image processing and computer vision techniques to identify and count individual maize plants in the aerial imagery. The development of a specialized algorithm tailored to maize plants contributes to the accuracy and efficiency of plant counting tasks.

- Validation against ground-truth data: The proposed counting algorithm is validated against ground-truth data obtained through manual counting in the field. This validation process ensures the accuracy and reliability of the digital counting approach, providing evidence of its performance in comparison to traditional manual counting methods.

2.5.3 Weakness

- Dependency on specific maize field conditions: The paper may not address the potential limitations associated with specific field conditions or variations in maize plant growth. Different factors, such as plant density, plant height, or variations in crop health, can affect the accuracy of plant counting. The paper may lack a detailed analysis of the system's performance in diverse field conditions or the robustness of the counting algorithm to these variations.

- Generalizability to other crop types: While the paper focuses on counting maize plants, it may not explicitly discuss the generalizability of the proposed approach to other crop types. Different crops may have distinct characteristics, growth patterns, or plant arrangements that may require modifications or adaptations to the counting algorithm. Further exploration and evaluation across multiple crop types would enhance the applicability and relevance of the proposed method.

- Potential for data processing challenges: The paper may not extensively discuss the potential challenges related to data processing and analysis of the acquired aerial imagery. Handling large volumes of high-resolution imagery can require significant computational resources and time. The paper may lack detailed insights into the data processing pipeline, including methods for image stitching, storage, and analysis, as well as potential limitations or bottlenecks in these processes.

- Limited analysis of error sources: The paper may not provide a comprehensive analysis of the potential sources of errors in the digital counting approach. Factors such as image distortion, image noise, or errors in ground-truth data collection can impact the accuracy of the counting results. A thorough investigation and discussion of these error sources would contribute to a more robust evaluation of the proposed system.

2.6 DeepFruits: A Fruit Detection System Using Deep Neural Networks

2.6.1 Description

Accurate and efficient fruit detection is essential for various agricultural applications, including yield estimation, quality assessment, and automated harvesting. In this paper [11] a robust fruit detection system has been presented that leverages deep neural networks for accurate and real-time fruit detection in agricultural settings.

The DeepFruits system employs state-of-the-art deep learning techniques to detect and localize fruits in images captured by vision sensors or cameras. A large-scale fruit dataset is constructed comprising diverse fruit types, varying lighting conditions, occlusions, and complex backgrounds. This dataset is used to train a deep convolutional neural network (CNN) to learn discriminative features and spatial relationships necessary for accurate fruit detection.

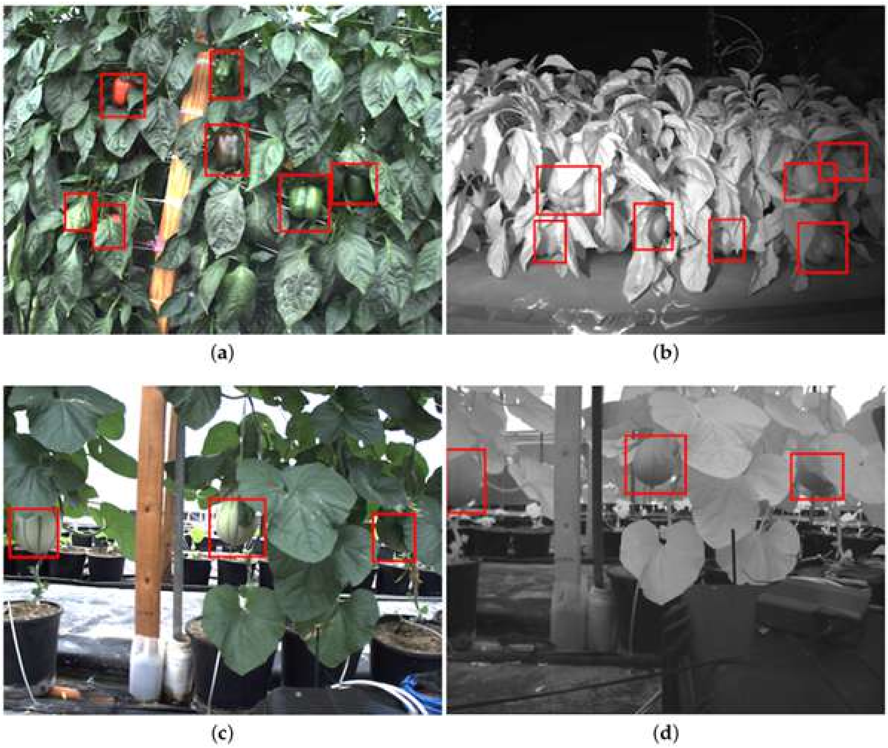


Figure 2.6: Example images of the detection for two fruits [11]

The trained CNN effectively detects fruits by analyzing the input images and generating bounding boxes around the fruit instances. The detection process is performed in a single-pass manner, enabling real-time operation and scalability. Furthermore, the CNN is designed to handle various fruit sizes, shapes, and orientations, ensuring robust performance across different fruit species.

To evaluate the effectiveness of the DeepFruits system, extensive experiments are conducted on multiple datasets captured in real agricultural environments. The results demonstrate the superior performance of this system, achieving high precision and recall rates across various fruit types. The DeepFruits system outperforms traditional computer vision methods and showcases its ability to handle challenging scenarios, including occlusions, overlapping fruits, and varying lighting conditions.

Additionally, the real-time performance of the DeepFruits system is evaluated on different hardware platforms. The system demonstrates efficient execution and minimal latency, making it suitable for deployment on embedded systems or edge devices for on-site fruit detection applications.

The DeepFruits system offers numerous benefits to the agricultural industry. By automating fruit detection, farmers can streamline crop monitoring, optimize resource allocation, and improve overall productivity. The system's accuracy and efficiency contribute to reducing labor costs and enhancing the quality and consistency of fruit-related decisions. Moreover, the scalability and adaptability of the system make it suitable for large-scale fruit detection applications, ranging from orchards to greenhouse environments.

2.6.2 Strength

- The paper introduces a novel approach for fruit detection in agricultural settings using deep neural networks. By leveraging the power of deep learning techniques, the proposed system addresses the challenges associated with accurate and real-time fruit detection, offering a fresh and innovative perspective on the topic.

- The paper leverages state-of-the-art deep neural networks, specifically deep convolutional neural networks (CNNs), for fruit detection. By employing deep learning, the system learns discriminative features and spatial relationships necessary for accurate fruit detection, surpassing traditional computer vision methods and showcasing the power of deep neural networks in addressing complex agricultural tasks.

- The paper demonstrates the effectiveness of the DeepFruits system through extensive experiments and evaluation on multiple datasets captured in real agricultural environments. The system achieves high precision and recall rates across various fruit types, outperforming traditional methods and showcasing its robustness in handling challenging scenarios, such as occlusions and varying lighting conditions. Much quicker to deploy and improved accuracy with 0.84 F1 score for sweet pepper.

- The DeepFruits system is designed to provide real-time fruit detection, making it applicable for practical deployment in agricultural settings. The system's accuracy, efficiency, and scalability contribute to enhanced crop management practices, automated fruit monitoring, and improved decision-making for farmers and researchers.

2.6.3 Weakness

- Limited diversity in fruit types: The paper may have limitations in terms of the variety of fruit types covered by the DeepFruits system. The effectiveness and generalizability of the proposed approach for fruit detection may vary across different fruit species with distinct visual characteristics. A broader representation of fruit types in the dataset and evaluation would enhance the applicability of the system to a wider range of fruits.

- Limited analysis of computational requirements: The paper may not provide a detailed analysis of the computational requirements of the DeepFruits system. Deep neural networks can be computationally demanding, especially during training and inference stages. The paper may lack insights into the computational efficiency, model size, and hardware requirements necessary to deploy the system in real-world applications, potentially limiting its scalability and practical feasibility.

- Limited evaluation in diverse environments: The paper may not thoroughly evaluate the performance of the DeepFruits system in diverse environmental conditions. Fruit detection and segmentation can be influenced by variations in lighting, backgrounds, occlusions, or growth stages. An evaluation across different environmental settings would provide a more comprehensive understanding of the system's robustness and generalizability in real-world scenarios.

- Lack of comparison with alternative methods: The paper may not include a comprehensive comparison of the DeepFruits system with alternative fruit detection methods or approaches. Comparative analysis against other computer vision techniques or state-of-the-art fruit detection systems would provide insights into the system's performance, strengths, and potential areas for improvement in relation to existing solutions.

2.7 Rapeseed Seedling Stand Counting and Seeding Performance Evaluation at Two Early Growth Stages Based on Unmanned Aerial Vehicle Imagery

2.7.1 Description

Accurate and timely assessment of seedling stand count and seeding performance is crucial for effective crop management, particularly in the early stages of growth. In this paper [12], a novel approach is presented for rapeseed seedling stand counting and seeding performance evaluation based on imagery captured by Unmanned Aerial Vehicles (UAVs).

This method leverages high-resolution aerial imagery acquired by UAVs to analyze and quantify the number of rapeseed seedlings in agricultural fields. A comprehensive image processing pipeline is proposed that combines computer vision techniques and machine learning algorithms to automate the counting process and evaluate the performance of seeding operations.

The pipeline consists of multiple stages. First, the UAV imagery is preprocessed to enhance the quality and visibility of the rapeseed seedlings. This includes image rectification, normalization, and noise reduction techniques to ensure accurate detection and counting.

Next, there employed a state-of-the-art computer vision algorithms to detect and segment individual rapeseed seedlings within the aerial images. Techniques such as edge detection, clustering, and region-based segmentation are utilized to identify and delineate seedling instances. The segmentation results are refined using morphological operations and geometric constraints to improve accuracy.

To evaluate the seeding performance, the detected seedling stand count is compared with the expected seeding rate or target stand count. Deviations from the target count indicate variations in seed distribution, germination rate, or plant emergence, providing insights into the effectiveness of the seeding process.

This approach is validated by conducting field experiments and collecting UAV imagery at two early growth stages of rapeseed. The results demonstrate the efficacy of this method in accurately estimating the seedling stand count and assessing seeding performance. The system achieves high accuracy compared to manual counting methods, reducing time and labor costs associated with traditional ground-based assessments.

The proposed approach offers significant advantages for rapeseed growers and researchers. It enables rapid and non-destructive monitoring of seedling stand count, facilitating early interventions for crop management. The evaluation of seeding performance helps identify potential issues, such as seed quality, seeding equipment efficiency, or environmental factors affecting germination and emergence.

2.7.2 Strength

- Use of unmanned aerial vehicles (UAVs) and aerial imagery: The paper leverages the use of UAVs to capture aerial imagery of rapeseed fields at two early growth stages. UAVs offer a unique vantage point for data collection, allowing for efficient and comprehensive coverage of large agricultural areas.

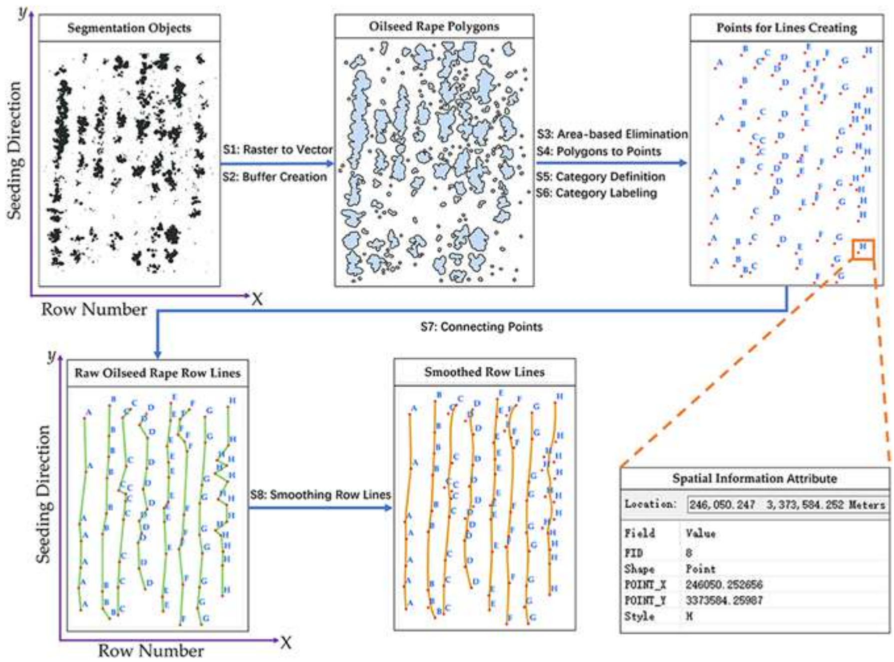


Figure 2.7: Flowchart for creating rapeseed row lines [12]

The utilization of aerial imagery provides valuable information for rapeseed seedling stand counting and performance evaluation.

- Focus on rapeseed seedling stand counting: The paper addresses the specific task of rapeseed seedling stand counting, which is crucial for assessing crop establishment and evaluating seeding performance. Accurate stand counts enable farmers and researchers to make informed decisions regarding seeding rates, plant populations, and overall crop health. The paper's emphasis on this important aspect contributes to the practical relevance of the research.

- Evaluation at two early growth stages: The paper examines rapeseed seedling stand counting and seeding performance at two critical early growth stages. Assessing the performance and accuracy of counting methods at different growth stages provides insights into the applicability and robustness of the proposed approach throughout the crop's development. It allows for a comprehensive understanding of the counting system's performance in dynamic agricultural environments.

- Integration of image analysis algorithms: The paper presents the development and integration of image analysis algorithms specifically tailored for rapeseed seedling stand counting. These algorithms utilize advanced computer vision techniques to identify and count individual seedlings in the aerial imagery. The incorporation of specialized algorithms enhances the accuracy and efficiency of the counting process.

- Validation against ground-truth data: The proposed counting method is validated against ground-truth data obtained through manual counting in the field. This validation process ensures the accuracy and reliability of the counting approach, providing evidence of its performance in comparison to traditional manual counting methods. The validation adds credibility to the proposed method and its suitability for practical use.

2.7.3 Weakness

- Limited generalizability: The paper may not extensively discuss the generalizability of the proposed method to different growing conditions, regions, or crop varieties. Rapeseed growth and appearance can vary based on factors such as soil type, climate, and management practices. The lack of analysis on the method's performance across diverse environments may limit its applicability and reliability in practical agricultural settings beyond the specific conditions examined in the study.

- Reliance on manual ground-truth data: The paper relies on manual counting as the ground-truth data for validating the seedling stand counting approach. Manual counting can be subjective and prone to human error, leading to potential discrepancies in the validation process. Further investigation into the accuracy and reliability of manual counting methods, as well as potential sources of error, would enhance the credibility of the results.

- Lack of analysis on varying seedling densities: The paper may not thoroughly analyze the performance of the counting method at different seedling densities. Varying seedling densities can pose challenges for accurate counting, especially when seedlings are closely spaced or overlapping. Evaluating the system's performance under different density conditions would provide a more comprehensive understanding of its robustness and limitations.

- Absence of comparative analysis: The paper may not include a thorough comparison of the proposed method with alternative approaches or existing seedling stand counting methods. Comparative analysis against other image analysis techniques or computer vision algorithms would provide insights into the strengths, weaknesses, and potential improvements of the proposed method in relation to existing solutions.

Working on 17509 images for the purpose of weed detection, some researchers in the paper "DeepWeeds: A Multiclass Weed Species Image Dataset for Deep Learning" [6] showed 95.1% and 95.7% accuracy using Inception-v3 [29] and ResNet-50 [33] models, respectively. For eight different weed species, this was a multiclass-classification problem. This project uses transfer learning to train their model. Image datasets were labeled, which facilitated the training phase.

Using Faster R-CNN and Kalman filter, the paper [7] attempted to count cotton seedlings in the early stage. The work was a CNN-based [28] approach. Constructing 3D data points from 2D images was a very challenging task that they dealt with in a smart way via the advanced video tracking algorithms (Kalman filter and optical flow). With a better F1 score (0.727) and R squared value (0.98), the model performs well for seedling detection. In addition to cotton seedling detection, the model can also differentiate weeds from seedlings.

For counting maize plants and calculating density, the paper [10] applied green pixel segmentation. By using thresholding, it can almost detect all the seedlings. But trouble arises when blurriness and weeds are present in the image. That time, the model leads to miscounting. Ultimately, 90% counting accuracy has been achieved.

In our findings, a fruit detection model [11] seems enticing for flawlessly detecting fruits in the tree. RGB and NIR multi-modal fused images have been used. Early fusion (feature-level fusion) and late fusion (decision-level fusion) methods both have limitations, like overfitting in the case of early fusion and suboptimal performance for late fusion. Here, the combination of the two decreases the constraint and increases the f1 score (0.838). Due to it being a light model, it is very simple to deploy the model. But the model is trained with small datasets. That's why it is not applicable to all the conditions.

We are concentrating on building a deep learning model that combines the two different types of work of seedling and weed detection. Therefore, we are now focusing on enhancing the existing methods. Just as it is important to keep the plant density equal throughout the land for the maximum use of the farmland, it is necessary to apply fertilizers and pesticides by detecting the weeds at an early stage to ensure the growth of the plant and proper nutrition. If accuracy can be confirmed, in that case, an overlapping framework will provide overall better performance.

Chapter 3

Methodology

3.1 Evaluation metrices

Evaluation metrics provide quantitative measures to assess how well a deep learning model is performing on a given task. It enable the comparison of different models or variations of a model. Metrics help diagnose issues such as overfitting or underfitting. Before deploying a deep learning model in real-world scenarios, it's crucial to assess its performance using relevant metrics. So in this section, the definition and mathematical formula of all the metrics by which our results can be judged are highlighted.

1. Accuracy: Accuracy is a fundamental performance metric that measures the overall correctness of a model's predictions across all classes. It is defined as the ratio of the number of correct predictions to the total number of predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.1}$$

where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

2. Precision: Precision is a performance metric that measures the accuracy of the positive predictions made by a model. It's a measure of a classifier's exactness. It is particularly relevant in binary and multiclass classification

problems.

$$Precision = \frac{TP}{TP + FP} \tag{3.2}$$

3. Recall (Sensitivity): Recall measures the percentage of actual positives that are correctly classified. It's a measure of a classifier's completeness. Recall is particularly relevant in scenarios where the cost of false negatives (missing a positive instance) is high

$$Recall = \frac{TP}{TP + FN} \tag{3.3}$$

4. F1 Score: The F1 score is the harmonic mean of precision and recall. It is particularly useful in scenarios where there is an uneven class distribution, and the trade-off between precision and recall needs to be considered.

$$\text{F1-score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3.4}$$

5. mAP (mean Average Precision): For object detection, Average Precision (AP) is calculated first. AP summarizes the shape of the precision/recall curve, and it's computed as the average precision at a set of eleven recall levels [0, 0.1, 0.2, ..., 1]. The mAP is the mean AP over all classes and/or over all images.

6. segm_mAP (mean Average Precision for segmentation): Each instance is evaluated separately, and the segmentation masks are compared with ground truth masks. segm_mAP specifically indicates that the mean Average Precision is calculated for the segmentation aspect of instance segmentation. It emphasizes the importance of evaluating the accuracy of the pixel-wise segmentation masks.

3.2 Classification task

Before collecting our own datasets, we worked with two different datasets that we got online. DeepWeeds [6] dataset was one of them. With this dataset, we trained five different classification models. All models are well known for classification tasks in deep learning. First, the dataset has been presented in Table 3.1, and next, a concise discussion of the architecture of the models is given.

Data	Count
Total number of images	17509
Training images	10502
Validation images	3501
Testing images	3507
Total number of classes	9

Table 3.1: DeepWeeds dataset summary

The dataset consists of 17509 images, collected from the Australian rangelands, with 9 labels (8 different weed classes and a negative class). The weed classes are Chinee apple, Lantana, Parkinsonia, Parthenium, Prickly acacia, Rubber vine, Siam wood, and Snake weed. With the image dataset, we got three CSV files (train_subset0.csv, val_subset0.csv, test_subset0.csv). Each CSV file has two columns, one for the filename with extension and another for the corresponding label (from 0 to 9). Training, validation, and testing datasets are divided into 60%, 20% and 20% of total images. The code was written in pytorch and run in Google Colab which is a free, cloud-based platform provided by Google that allows users to write and execute Python code in a collaborative environment. Colab provides free access to Graphics Processing Units (GPUs), in our case an Nvidia Tesla T4 GPU with maximum memory capacity of 15360MiB. It took more than three hours to train all the models.

1. SqueezeNet:

    - SqueezeNet [34] is a compact CNN architecture designed to be efficient in terms of model size and computation.
    - It achieves this efficiency by using a combination of 1x1 convolutions and "fire modules" that reduce the number of parameters while maintaining good accuracy.
    - SqueezeNet is suitable for resource-constrained devices and real-time applications.

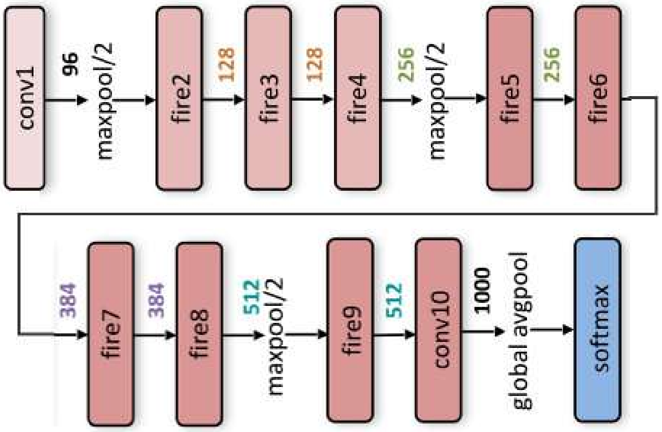


Figure 3.1: Fire module in SqueezeNet [13]

2. ShuffleNet:

    - ShuffleNet [14] focuses on reducing computation and memory usage by employing group-wise pointwise convolutions and channel shuffling techniques.
    - It divides channels into groups and shuffles them, allowing information to flow more efficiently and reducing parameters.
    - ShuffleNet is known for its efficiency and is suitable for tasks where computational resources are limited.

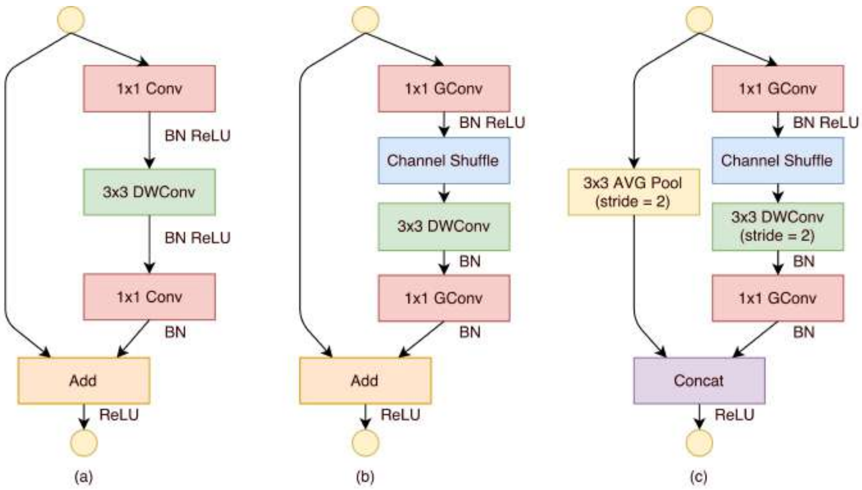


Figure 3.2: The architecture utilizes two new operations, pointwise group convolution and channel shuffle [14].

3. EfficientNet:

- EfficientNet [35] is a family of CNN architectures optimized for balancing model size, depth, and width.
- It uses a neural architecture search (NAS) approach to find the best trade-off between model complexity and performance.
- EfficientNet models achieve state-of-the-art results on various image classification tasks.

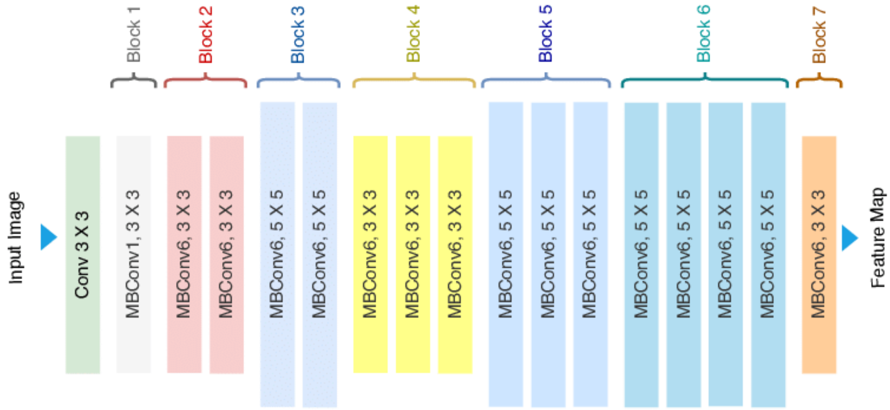


Figure 3.3: EfficientNet-B0 architecture [15]

4. DenseNet (Densely Connected Convolutional Networks):

- DenseNet [36] connects each layer to every other layer in a feed-forward fashion, facilitating feature reuse and enhancing gradient flow.
- It reduces vanishing gradient problems and leads to efficient parameter usage.
- DenseNet is known for its impressive performance and has applications in image classification and feature extraction.

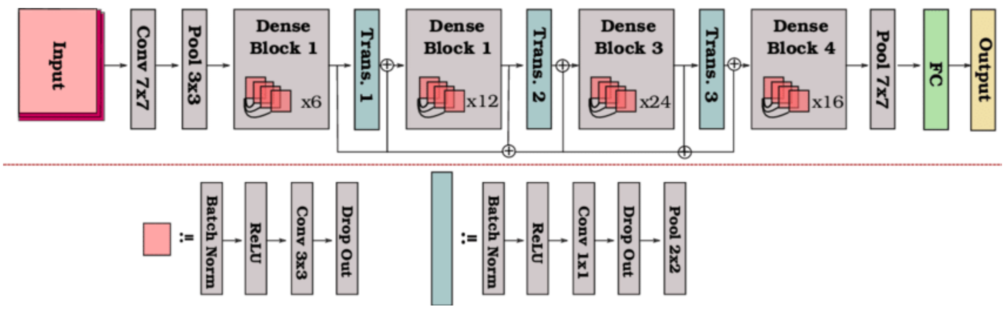


Figure 3.4: A schematic illustration of the DenseNet-121 architecture [16]

5. MobileNet:

- MobileNet [37] is designed for efficient inference on mobile and embedded devices.
- It uses depthwise separable convolutions to reduce computation and model size while maintaining accuracy.
- MobileNet is commonly used for real-time tasks like object detection and image segmentation on mobile platforms.

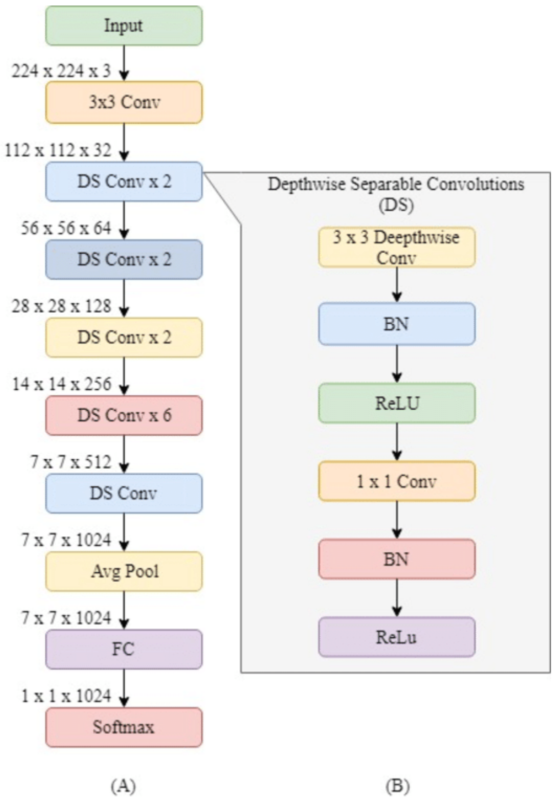


Figure 3.5: Illustration of the MobileNet architecture. (A) The overall MobileNet architecture and (B) an in-depth explanation of the DS layer [17]

3.3 Object detection task

The primary goal of our project was seedling segmentation. Although, later we worked on seedling segmentation. Section 5 will cover the methodology of seedling segmentation. Two different datasets (one of which we collected ourselves) were the main resources of our object detection task. DeepSeedling dataset includes three image datasets used for plant seedling detection. They referred them as TAMU2015, UGA2015, and UGA2018. We worked on only TAMU2015 dataset, so the description of the dataset is given in Table 3.2.

Data	Count
Total number of images	10921
Annotated images	1820
Annotated training images	1456
Annotated validation images	182
Annotated testing images	182
Total number of classes (Without background)	2

Table 3.2: TAMU2015 Dataset Summary

TAMU2015 consists of 10921 drone images, with 2 classes (Plant, Weed), but only 1820 images are annotated in json format. The JSON annotation format of the dataset is slightly different which was not working after uploading to Roboflow. So at first, we separated the 1820 specific images from 10921 images. Then we made a coco json annotation file writing a python script which became supported by Roboflow. The annotation file was created following the original annotation file. Next we uploaded the images and corresponding annotation in Roboflow. By default, Roboflow annotated the images reading the annotation file. But we need to correct some bounding box in the images since there were so many flaws in annotations. After generating a version of our dataset which were images of size 640 by 640 with proper augmentation (horizontal flip, slightly blurred, slightly increased brightness, slightly cropped, slightly angled). We split the dataset into train, valid and test into 80%, 10% and 10%. After preprocessing the dataset, we called Roboflow account's API in Google Colab in code section to download the generated dataset.

Roboflow provides its own YOLOv8 code frame to ease the training, validation and testing process. Before training we tweaked the architecture of YOLOv8. Finally We trained the dataset up to 100 epochs. The architecture of YOLOv8 model is shown in the Fig. 3.6.

YOLOv8:

- Latest model in the YOLO family, introduced in 2022 by Ultralytics.
- Built on the YOLOv5 framework and includes several architectural and developer experience improvements.
- Faster and more accurate than YOLOv5.
- Provides a unified framework for training models for performing object detection, instance segmentation, and image classification.
- Boasts an advanced training scheme with knowledge distillation and pseudo-labeling.

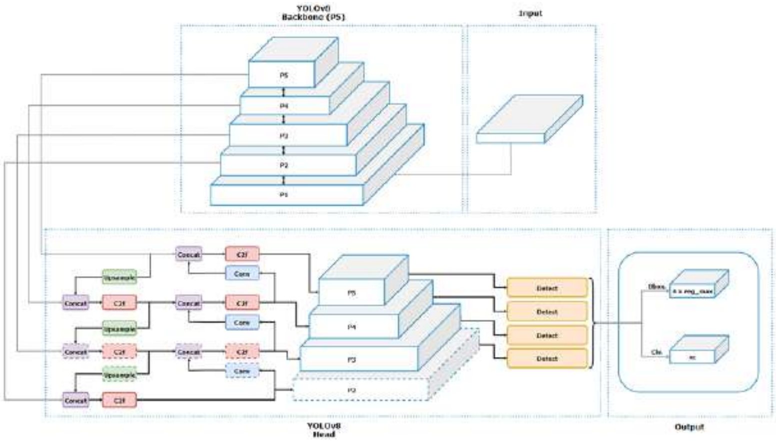


Figure 3.6: The improved YOLOv8 network architecture [18]

3.3.1 Roboflow:

Roboflow is a platform that provides tools and services for working with computer vision datasets. It facilitates the annotation and labeling of images for computer vision tasks. Users can annotate objects in images and define classes to create labeled datasets. The platform supports various data augmentation techniques to increase the diversity of the training dataset. It provides tools for versioning datasets, allowing users to track changes over time and maintain a history of different dataset versions. Once the datasets are prepared and augmented, Roboflow allows users to export the datasets in formats compatible with various machine learning frameworks. The platform also includes collaboration features, enabling teams to work together on dataset preparation. Users can share datasets with collaborators and manage access rights.

3.4 Dataset Collection

3.4.1 Study Area

The experiment was conducted at the BSMRAU experiment field that is shown in Fig. 3.7. The experiment was laid on a completely randomized block design (RCBD). Final land preparation, including basal dose application and soil treatment, was done on 15 July 2023. The procedures, from land preparation to weeding and thinning, have been shown datewise in Table 3.3. Cotton seeds were collected from the Bangladesh Cotton Development Board. Two types of cotton seeds (high-yielding and hybrid) were sown one week after the final land preparation. Cotton germination was closely monitored. Subsequently, gaps were filled to maintain uniform plant-to-plant and row-to-row distances. Other intercultural operations like weeding and thinning were also conducted.

Operation	Date
Land Preparation	15 July 2023
Sowing date	22 July 2023
Start of germination	26 July 2023
Germination completed	30 July 2023
Gap filling	02 August 2023
Thinning and weeding	21 August 2023
Experimental design	RCBD
Plot size	$7.30m^2$

Table 3.3: Land Preparation for the Study

3.4.2 Data Acquisition

Cotton plant photos were taken using a low-cost DJI mini2 drone when they attended a 12-15 cm height. Drone photos were taken just above the canopy (nadir) at a height of 1.5 meters. Camera descriptions are detailed in Table 3.4. It

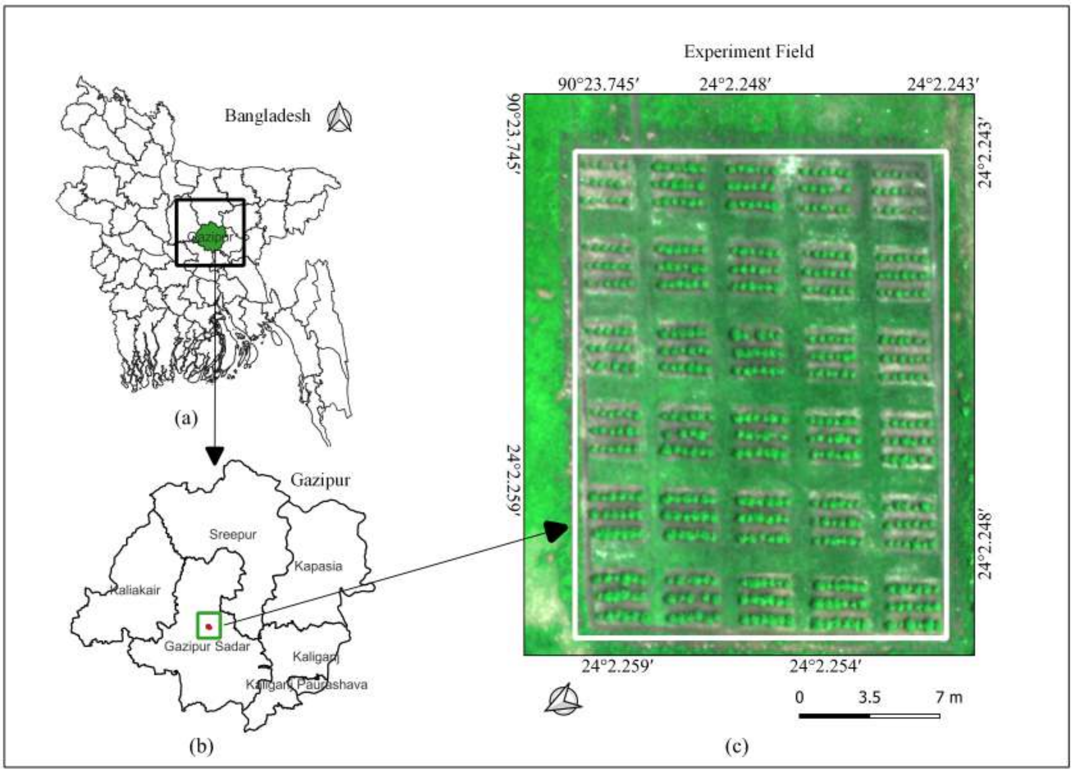


Figure 3.7: Study Area Map of Cotton Field

was equipped with a 1/2.3” CMOS sensor that offers an effective pixel count of 12 megapixels, ensuring high-resolution images. For image stabilization, it employed a 3-axis system which includes tilt, roll, and pan adjustments. The lens had a field of view (FOV) of 83 degrees and comes with an aperture setting of f/2.8. This camera captured photos in JPEG format. The images of the seedlings were taken from a height of 5 meters above the canopy. The shutter speed of the camera is quite versatile, with an electronic shutter range from 4 seconds to 1/8000 of a second. Lastly, the maximum image size that the camera can capture is in two aspect ratios: 4:3 with a resolution of 4000x3000 pixels, and 16:9 with a resolution of 4000x2250 pixels, allowing for detailed images.

Camera description	DJI mini2
Sensor	1/2.3" CMOS Effective Pixels: 12 MP
Stabilization	3-axis (tilt, roll, pan)
Lens	FOV: 83° with aperture f/2.8
Photo Formats	JPEG
Photo taken	5 meters above canopy
Shutter Speed	Electronic Shutter: 4-1/8000 s
Max Image Size	4:3: 4000×3000
	16:9: 4000×2250

Table 3.4: UAV RGB Camera Details

Figure 3.8: Seedling image of our dataset

Figure 3.9: Seedling image of our dataset

3.4.3 Dataset Description

Crop parameter	Information
Scientific name	Gossypium hirsutum
Hybrid variety	CB Hybrid-1
High yielding variety	CB-15
Source of seed	Cotton Development Board
Seedling height	<15 cm

Table 3.5: Crop Information

The scientific name of the crop is Gossypium hirsutum. It is type of cotton. The crop has two varities, a hybride variety (CB Hybrid-1) and a high yeilding variety (CB-15) denoting a focus on both productivity and potential cross-breeding benefits. The seeds come from the Cotton Development Board that is a quality-controlled source. Seedling height is currently under 15 cm, indicating an early growth stage.

3.4.4 Image Pre-Processing

The images we collected for our project were of very high resolution, measuring 12000 by 9000 pixels. This high resolution presented a challenge during the annotation process, as the cotton seedlings appeared very small when viewed at 1x zoom, making them difficult to accurately annotate. To address this challenge, we cropped different sections from the images to reduce the resolution and increase the dataset size. This also made the seedlings more visible. The resolution of the cropped images was not fixed, but after data augmentation, all images were resized to 640 by 640 pixels. Data augmentation will be described in more detail in the next section.

3.4.5 Data Annotation

3.4.5.1 Bounding box annotation

Images in our dataset were annotated using the Roboflow [38] platform. After creating a new project, the 'Object Detection' annotation type was selected.

Bounding boxes were drawn around each seedling in the images, labeling them as "Seedling". It took about three days to finish the annotation. Once annotations were completed, the dataset was exported. In total, 714 images were annotated. These images were then divided into training, validation, and testing sets at ratios of 70:20:10, ensuring the model would be assessed on unfamiliar images.

Figure 3.10: Box annotated image

3.4.5.2 Mask annotation

Later, we mask annotated the images as cascade mask-rcnn needs mask annotation to work with. Similar processes are followed as bounding box annotation. The difference is that mask annotation was done instead of bounding box annotation. Dataset was divided into 80:10:10 ratios for training, validation and testing purposes.

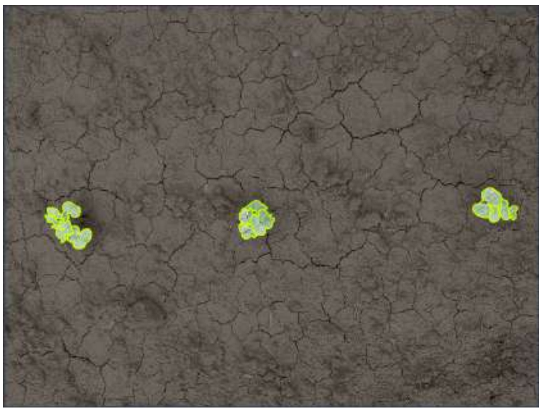


Figure 3.11: Mask annotated image

3.4.6 Data Augmentation

Several data augmentation techniques were applied to the sapling dataset to enhance its variety and robustness. These methods encompassed horizontal and vertical flips, 90-degree rotations, and cropping ranging from no zoom to a maximum of 50%. Additionally, images underwent rotations ranging between -15 and +15 degrees, horizontal and vertical shearing of up to 15 degrees, and modifications to saturation, brightness, and exposure, with variations from -25% to +25%. Other adjustments included blurring effects up to 1.25 pixels and the addition of noise affecting up to 5% of pixels.

Table 3.6: Number of images for train, val, and test at each experiment

IMAGES	DETECTION	SEGMENTATION
TRAIN	1500	571
VAL	142	71
TEST	72	72
TOTAL	1714	714

For seedling detection, post-augmentation dataset expanded to 1714 images. To train, validate, and test our model, the augmented images were divided in a 70:20:10 ratio. This split resulted in approximately 1500 training images, 142 for validation, and 72 for testing.

For seedling segmentation, after augmentation, the number of images in the dataset has not increased. Here, the dataset was split into an 80:10:10 ratio which generated 571, 71, and 72 images for training, testing, and validation, respectively.

3.5 Training with the dataset

3.5.1 Seedling detection

We have trained three models with this cotton seedling dataset. YOLO and its different versions are the state-of-the-art model that perform better in terms of object detection. Code samples are taken from Roboflow model [38] site. Some modification was needed to suit the code with the dataset. Training was done on Google Colab. YOLOv5 has been trained upto 100 epochs, likewise YOLOv8 up to 100 epochs, although YOLONAS up to 25 epochs (training time was longer in this case).

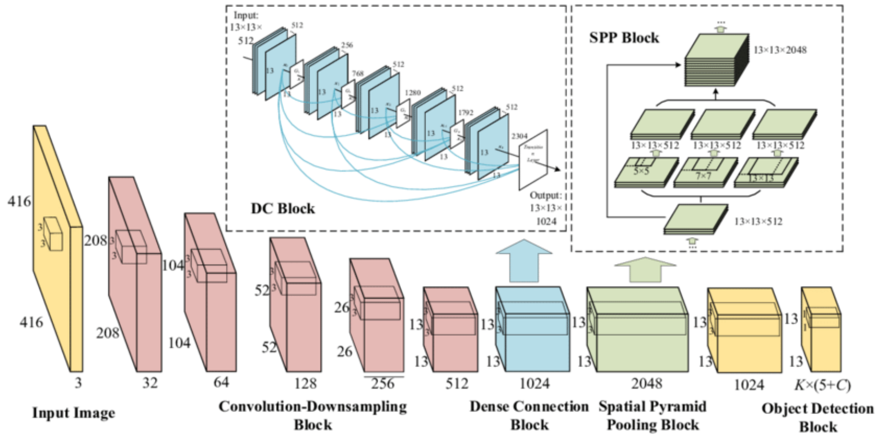


Figure 3.12: YOLO architecture in-depth [19]

- YOLOv5
  - Introduced in 2020 by Ultralytics, built on the PyTorch framework.
  - Fast, easy to use, and capable of achieving state-of-the-art results for object detection tasks.
  - More accurate and easier to train than its predecessors.
  - Popular choice for many developers.

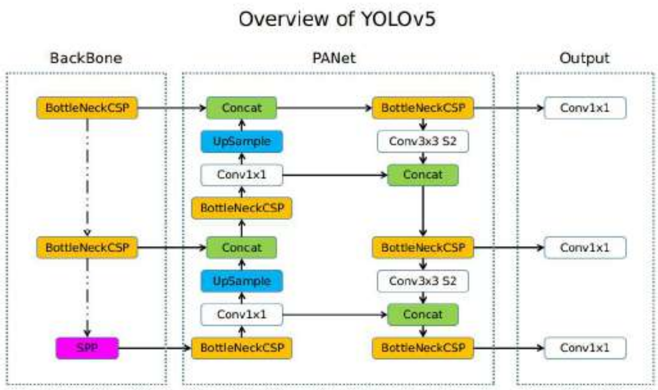


Figure 3.13: Overview of model structure about YOLOv5 [20]

- YOLO-NAS:
  - New real-time state-of-the-art object detection model introduced in 2023.
  - Outperforms both YOLOv6 and YOLOv8 models in terms of mAP (mean average precision).
  - Significantly improves small object detection, localization accuracy, and performance-per-compute ratio.
  - Ideal for real-time edge computing applications.
  - Better regarding the number of metrics than YOLOv8 and YOLOv6.

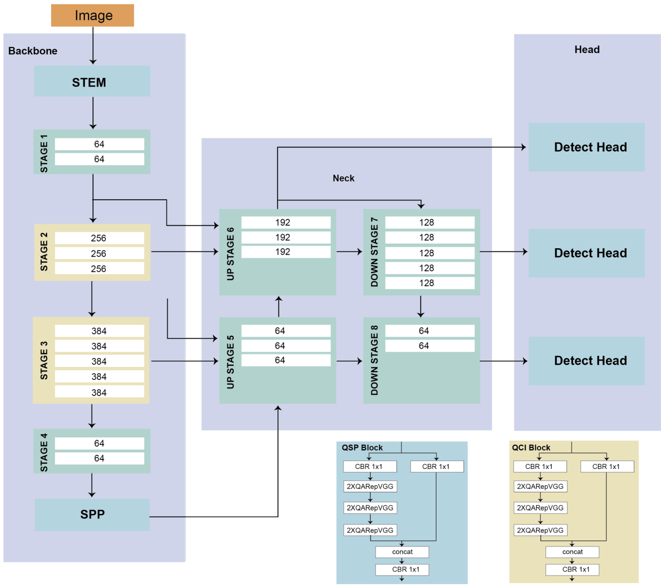


Figure 3.14: YOLO-NASL architecture

[39]

3.5.2 Seedling segmentation

Instance segmentation can be done by several different models like mask-rcnn, cascade mask-rcnn, panoptic fpn, yolact, detection transformer. Among them mask-rcnn is the most popular model in instance segmetation field. But we selected cascade mask-rcnn model due to its improved accuracy, robustness to false positives and fine-grained refinement than mask-rcnn. We have first trained using a default cascade mask-rcnn model named cascade-mask-rcnn_r50_fpn. Afterwards, we made some changes to the architecture of the model. We re-trained with this modified model. Cascade Mask R-CNN code sample was taken from mm detection toolbox. MM Detection, short for OpenMMLab Detection Toolbox, is an open-source software toolkit for object detection. MM Detection supports various state-of-the-art object detection models, including Faster R-CNN [32], Mask R-CNN [40], Cascade R-CNN [41], and more. The architecture of cascade mask-rcnn module is shown in Fig. 3.15. The dashed border delineates the two-stage detection phase of Cascade Mask R-CNN.

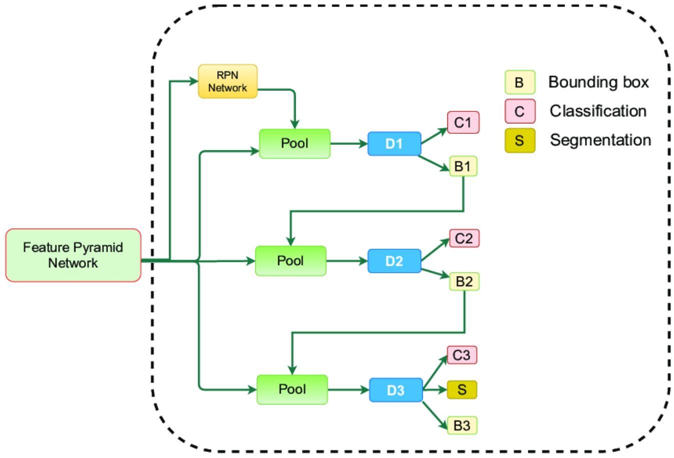


Figure 3.15: Explained architecture of Cascade Mask R-CNN module [21]

We have trained 4 models, two with default model and two with modified model but wih different batch size and epoch number. First three models were trained in PC with dedicated GPU. The last model was trained in Google Colab server due to memory shortage in the PC. A model for memory shortage is provided on Google Colab server. The configuration of hardware and software on PC and Google Colab were given below.

Table 3.7: Information about Hardware and Software experimental environment.

Name	Parameter
Experimenter environment	PC with GPU integration / Google colab
GPU	NVIDIA RTX 3060 Ti / NVIDIA Tesla T4
Deep learning framework	Torch-2.1.2+cu121 / Torch-2.1.0+cu121
Language	Python-3.8.18 / Python-3.10.12

The proposed modifications of the model:

1. Backbone: The modified model configuration includes a `dcn` (Deformable Convolutional Networks) module in the backbone. This module is introduced using the keys `dcn=dict(type='DCN', deform_groups=1, fallback_on_stride=False)` and `stage_with_dcn=(False, True, True, True)`. Here the deformable convolutions are being used in certain stages of the ResNet backbone.

2. Neck: An additional neck module, `BFP` (BiFPN) is added after the FPN and has `in_channels=256` and `num_levels=5`. That means we are using BiFPN as an additional feature pyramid network.

3. ROI Head: The `bbox_roi_extractor` in the ROI head is changed to `GenericRoIExtractor`, and it includes a `GeneralizedAttention` module in the `post_cfg`. It is a different approach to extracting features from RoIs, including attention mechanisms. The `bbox_head` in the ROI head has been changed from the original `Shared2FCBBoxHead` to `SABLHead`. The `SABLHead` has multiple differences in terms of its architecture, including the use of bucketing bounding box coding, different scale factors, and modifications in the number of convolutional layers, among other parameters.

4. Mask ROI Extractor and Head: Similar to the changes in the `bbox_roi_extractor`, the `mask_roi_extractor` in the second configuration is also changed to `GenericRoIExtractor` with a `GeneralizedAttention` module in the `post_cfg`

5. Training Configuration: The training configuration (`train_cfg`) has some changes in terms of RPN assignment thresholds (`neg_iou_thr`, `min_pos_iou`).

6. Testing Configuration: The testing configuration (`test_cfg`) has change just in the RCNN score threshold (`score_thr`).

Chapter 4

Experimental Results

4.1 Weed Classification

We have trained all the models up to 25 epochs at which the accuracy has been saturated. Comparison metrics is shown in Fig. 4.1

Model	Accuracy	Precision	Recall	F1_score
SqueezeNet	65.3%	94.2%	65.3%	74.7%
ShuffleNet	93.8%	98.6%	93.8%	95.8%
MobileNet	83.7%	97.5%	83.7%	88.9%
EfficientNet	88.2%	97.7%	88.2%	92.1%
DenseNet121	94.0%	98.7%	94.0%	96.1%

Table 4.1: Performance metrics comparison of different classification models

DenseNet121 performed best in this case. Training and validation accuracy (Fig. 4.2) along with training validation loss (Fig. 4.1) with respect to the number of epochs, unormalized (Fig 4.3) confusion matrix of DenseNet are shown here. We can clearly see, while increasing the number of epochs, training and validation loss are decreasing, as well as accuracy is increasing. The overall results are anticipated from a deep learning model.

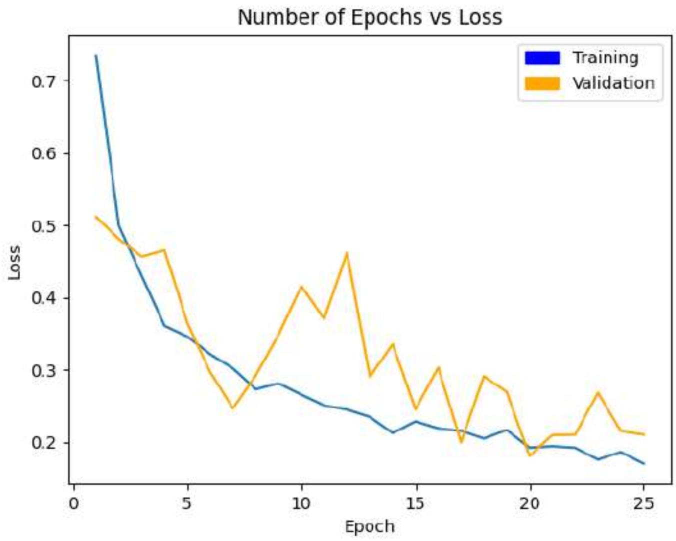


Figure 4.1: Training and Validation loss with respect to the number of epochs [DenseNet121]

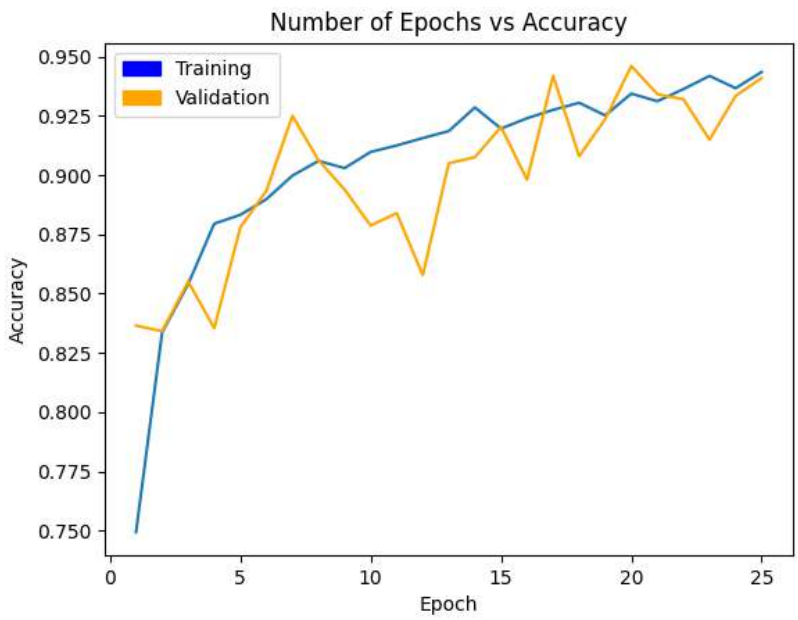


Figure 4.2: Training and Validation accuracy with respect to the number of epochs [DenseNet121]

During the training process, the accuracy of the model on the training dataset has increased nonlinearly, showing gradual improvement over time. However, when observing the validation accuracy, there have been some instances where a significant jump can be observed. Therefore, it is crucial to carefully monitor and analyze the validation accuracy to ensure that the model is not only performing well on the training data but also on new, unseen data.

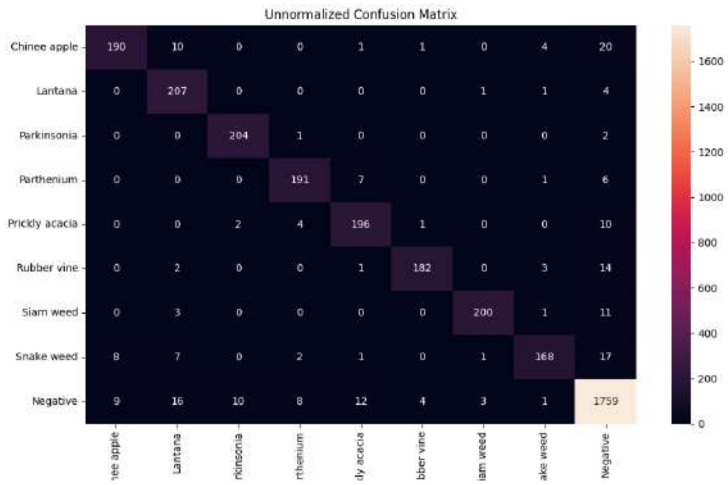


Figure 4.3: Confusion Matrix [DenseNet121]

From the confusion matrix, we can clearly see, the model (DenseNet121) performed better enough in classifying all nine classes. Only for the Chinee apple, the model performed relatively poor. The model showed best result in case of Parkinsonia. It predicted multiple cases as negative which are under false positive.

The author of the paper [6] trained their model applying transfer learning, found 95.1% and 95.7% accuracy on Inception-v3 and ResNet-50 respectively. Our results for all five models are shown in Chapter 4. Though the original code implementation was tensorflow-based, we wrote our code in pytorch.

4.2 Seedling detection on Deepseedling Dataset

We have trained upon the seedling dataset (TAMU2015) [7] using YOLOv8 [23] which is the latest model of "You Only Look Once" and found satisfactory results. Seedling detection with confidence score of our tuned YOLOv8 model as well as precision-recall table, precision-recall curve with compared to original paper are shown below.

Table 4.2: Deepseedling paper's [7] per-category performance of the model generalizability at the $IOU_{0.5}$

Training set	Testing set	PlantAP	PlantAR100	PlantF1	WeedAP	WeedAR100	WeedF1
T15U15 training	UGA2018 testing	0.981	1.000	0.0.991	NA	NA	NA
UGA2018 training	UGA2018 testing	0.983	0.992	0.987	NA	NA	NA
T15U18 training	UGA2015 testing	0.676	0.814	0.739	0.027	0.553	0.052
UGA2015 training	UGA2015 testing	0.988	0.995	0.991	0.859	1.000	0.924
U15U18 training	TAMU2015 testing	0.911	0.940	0.925	0.266	0.775	0.396
TAMU2015 training	TAMU2015 testing	0.999	0.999	0.999	0.980	1.000	0.990

Table 4.3: Testing Results using YOLOv8 model

Class	Images	Instances	Precision	Recall	mAP50	mAP50-90
all	100	1051	0.8652	0.953	0.943	0789
Plant	100	1037	0.963	0.977	0.988	0.903
Weed	100	14	0.762	0.929	0.898	0.675

Here in original paper's results table's, we are only concerned about the last row where findings of dataset TAMU2015 are shown. In terms of Plant and Weed classes results, our model's findings are quite close to original paper's findings. Their precision, recall values are 0.999 and 0.999 for plant and 1.000 and 0.990 for weed where our model's precision and recall values are 0.963 and 0.977 for plant and 0.762 and 0.929 for weed.

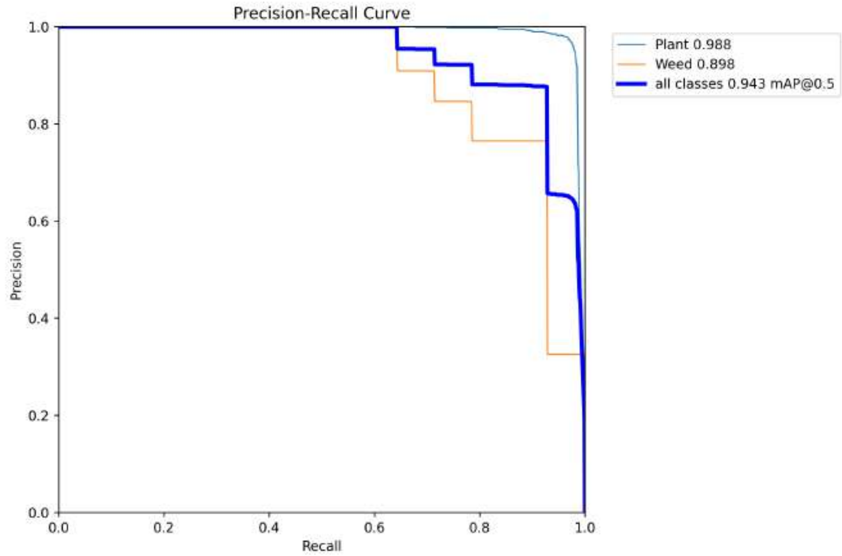


Figure 4.4: Precision-Recall curve using YOLOv8 model

A Precision-Recall (PR) curve is a graphical representation that plots the precision (y-axis) against the recall (x-axis) at various threshold settings.

In an ideal scenario, we would want a model to have both high precision and high recall, which means the model is able to correctly identify the positive cases (high precision) and doesn't miss a significant number of positive cases (high recall). This would be represented as a point in the top-right corner or a curve that skews towards the top-right corner of the plot.

Our model's prediction PR curve is gradually decreasing after recall value 0.67, it means that as we aim to increase the recall value (trying to identify more true positives), our model is mistakenly classifying more negative cases as positive, thereby reducing its precision. In other words, as we're trying to ensure we don't miss any positive cases (increasing recall), we end up with more false positives, decreasing the precision of the model.

This indicates a trade-off between precision and recall: increasing recall tends to decrease precision and vice versa. This trade-off is a common scenario in many real-world classification problems. The balance between precision and recall, and thus the shape and area under the PR curve, give a comprehensive measure of a model's performance.

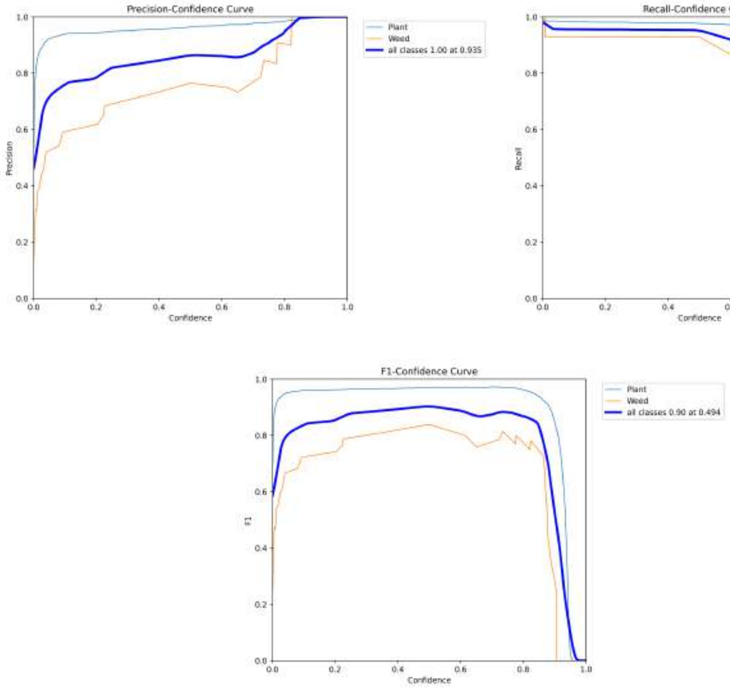


Figure 4.5: Precision (upper left), Recall (upper right) and F1 curve (bottom middle)

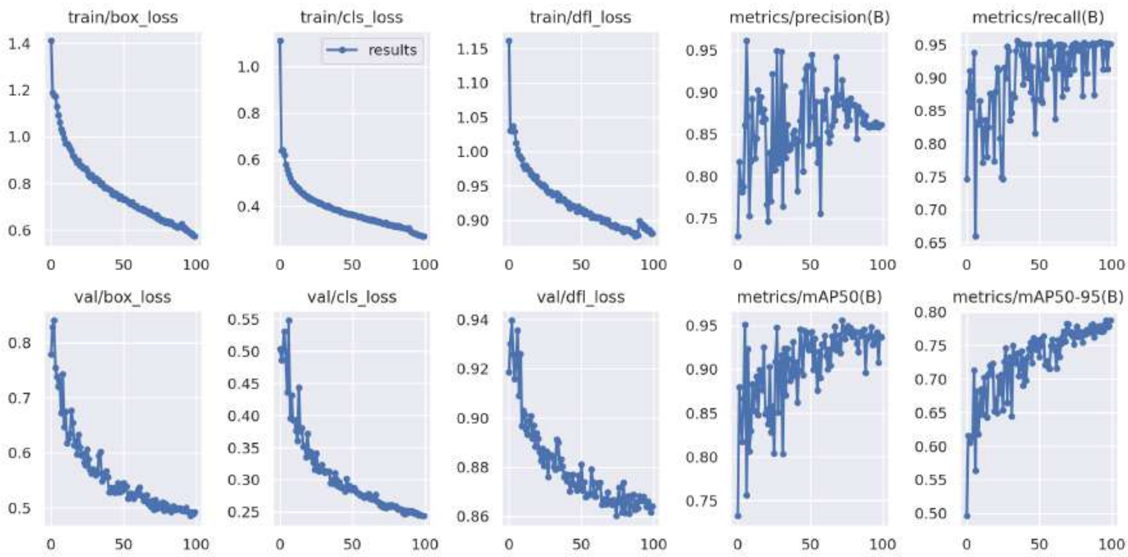


Figure 4.6: Different metrics in training phase

- Train/box_loss and val/box_loss measure the loss between the predicted bounding boxes and the ground truth bounding boxes. A lower box loss means that the model's predicted bounding boxes are more closely aligned with the ground truth bounding boxes.

- Train/cls_loss and val/cls_loss measure the loss between the predicted class probabilities and the ground truth class labels. A lower cls_loss means that the model's predicted class probabilities are more closely aligned with the ground truth class labels.

- Train/dfl_loss and val/dfl_loss measure the loss between the predicted feature maps and the ground truth feature maps. A lower dfl_loss means that the model's predicted feature maps are more closely aligned with the ground truth feature maps.

- Metrics/precision(B), metrics/recall(B), metrics/mAP50(B) are metrics that measure the model's performance on the validation dataset. Precision measures the fraction of predicted bounding boxes that are true positives. Recall measures the fraction of ground truth bounding boxes that are predicted as positives. mAP50 measures the mean average precision at an intersection over union (IOU) threshold of 0.5. mAP50-95 measures the mean average precision at an IOU threshold of 0.5:0.95.

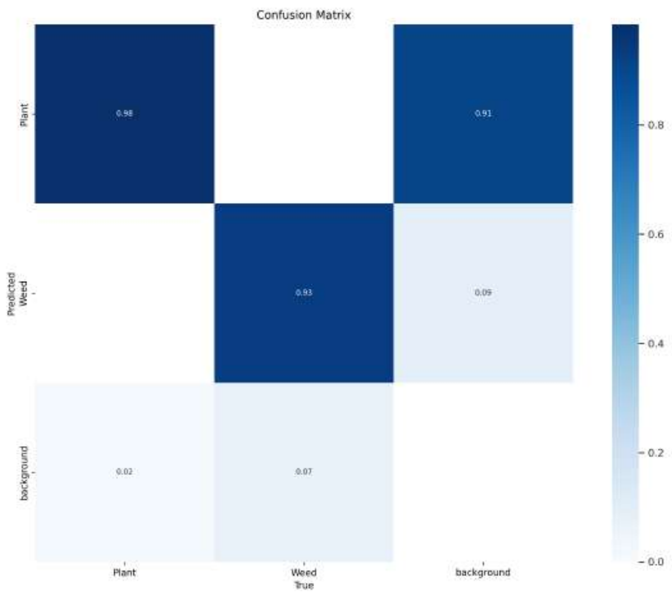


Figure 4.7: Confusion matrix of seedling and weed detection

From the confusion matrix, we can see here, our tuned YOLOv8 model successfully predicted plant and weed portion of an image. Here along the x-axis is indicated as true and along the y-axis is indicated as predicted. The rate of success for plant and weed detection are 98% and 93% respectfully.

A sample of some test and predicted images are shown here.

Figure 4.8: Input (left) and predicted (right) images

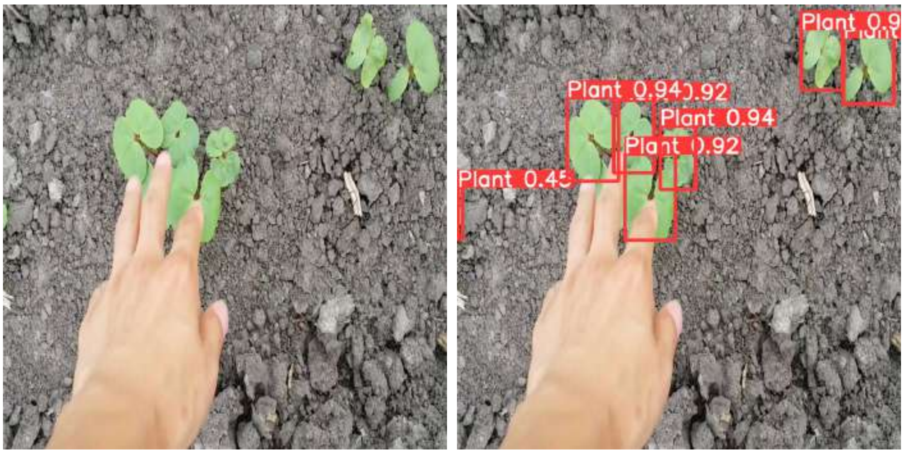


Figure 4.9: Input (left) and predicted (right) images

Figure 4.10: Input (left) and predicted (right) images

4.3 Seedling detection on Cotton Seedling Dataset

We have trained our Cotton Seedling Dataset using three different YOLO models (YOLOv5, YOLOv8 and YOLONAS).

Model	Precision	Recall	mAP@50
YOLOv5	85.6%	73.3%	80.4%
YOLOv8	91.1%	82.8%	91.1%
YOLONAS	7.29%	92.9%	81.8%

Table 4.4: Result comparison of different models

Performance Descriptions of Object Detection Models

4.3.1 YOLOv5:

- Precision: 85.6%
  This indicates that 85.6% of the detections made by YOLOv5 were relevant.
- Recall: 73.3%
  This suggests that YOLOv5 detected 73.3% of the total relevant instances present in the dataset.
- mAP@50: 80.4%
  The mean average precision at an Intersection over Union (IoU) threshold of 0.50 was 80.4%, meaning that, on average, 80.4% of the detections were accurate at this IoU level.

4.3.2 YOLOv8:

- Precision: 91.1%
  YOLOv8 had a precision of 91.1%, indicating a high rate of relevant detections.

- Recall: 82.8%

  This model detected 82.8% of all the relevant instances in the dataset.

- mAP@50: 91.1%

  The model had an mAP of 91.1% at an IoU threshold of 0.50, showcasing a high level of accuracy for the detected objects.

4.3.3 YOLONAS:

- Precision: 7.29%

  A significantly lower precision suggests that a large proportion of detections by YOLONAS were not relevant.

- Recall: 92.9%

  Despite its low precision, YOLONAS managed to detect 92.9% of all relevant instances, which is commendably high.

- mAP@50: 81.8%

  The mean average precision at the same IoU threshold was 81.8%, indicating a decent average accuracy across detections.

In summary, YOLOv8 appears to be the best-performing model among the three, achieving high precision, recall, and mAP@50. YOLOv5 also performs well, while YOLONAS, despite having a high recall, suffers from a significantly low precision.

Here the precision-recall curve is shown below-

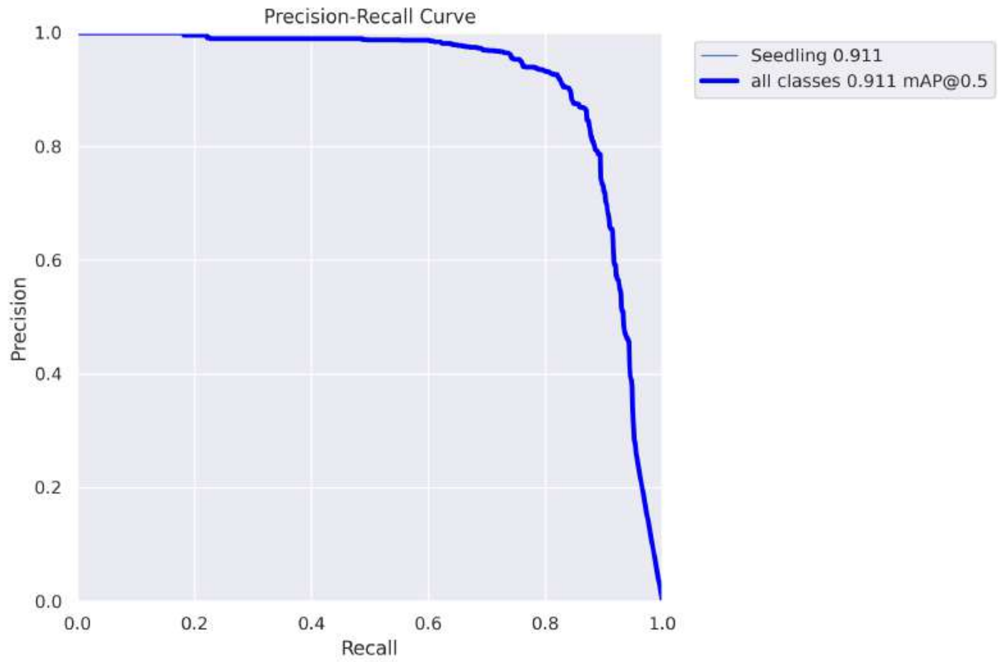


Figure 4.11: Precision-Recall curve of cotton seedling [YOLOv8]

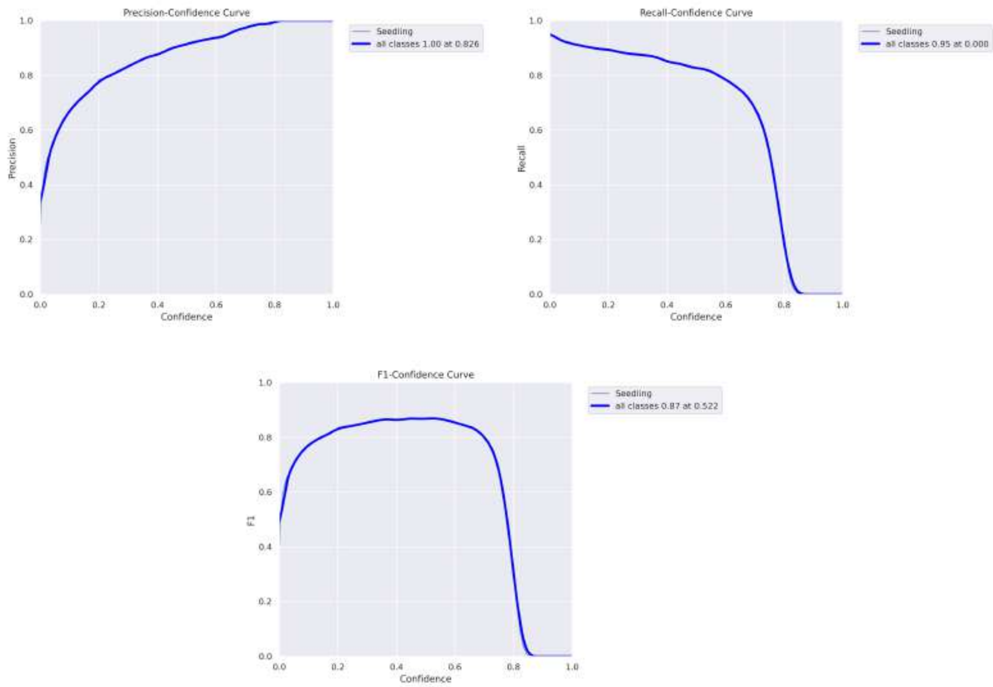


Figure 4.12: Precision (upper left), Recall (upper right) and F1 curve (bottom) of cotton seedling [YOLOv8]

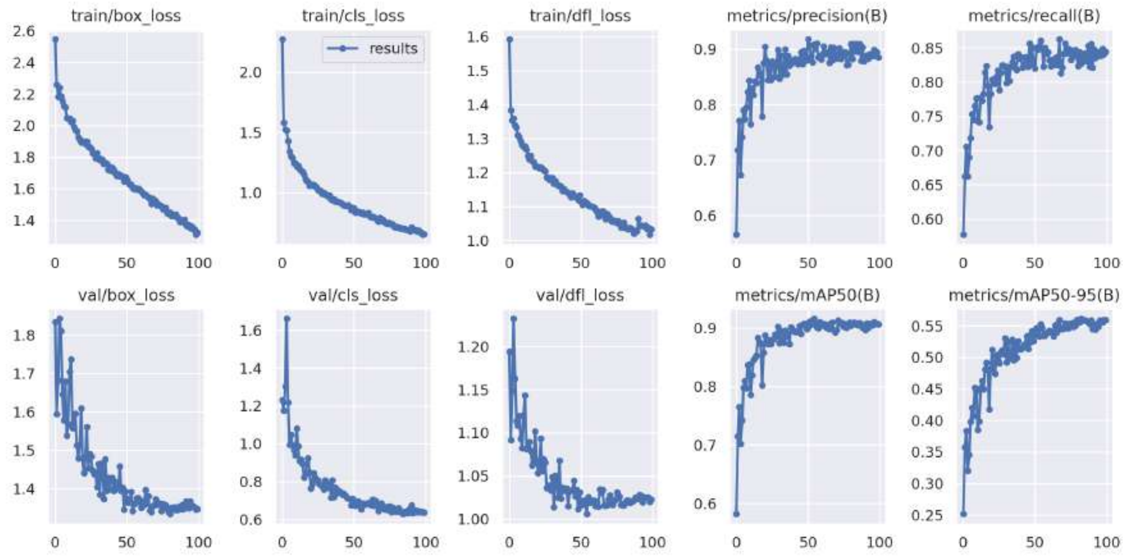


Figure 4.13: Different metrics in training phase of cotton seedling [YOLOv8]

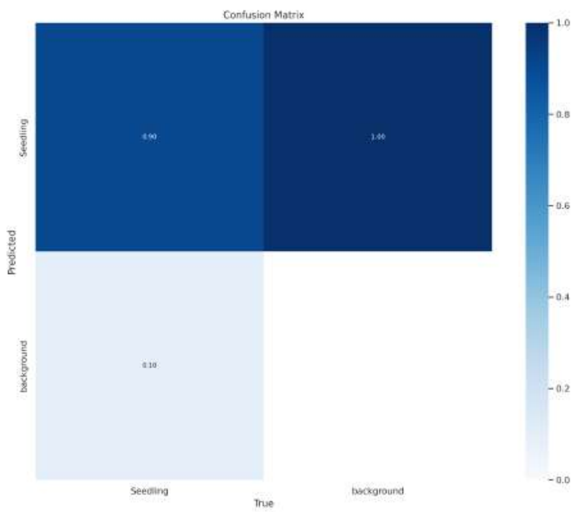


Figure 4.14: Confusion matrix of seedling detection [YOLOv8]

- True Positive (TP): When the actual class is seedling and the model also predicts seedling. This happens 90% of the time.
- False Negative (FN): When the actual class is seedling, but the model predicts background. This occurs 10% of the time.
- False Positive (FP): When the actual class is background, but the model predicts seedling. This happens 100% of the time.

- True Negative (TN): When the actual class is background and the model also predicts background. It appears this value is missing, but considering the nature of confusion matrices and the provided data, this value should be 0% (i.e., it never correctly identifies the background).

Based on this confusion matrix:

The model is pretty good at detecting seedlings when they are actually present, with a 90% accuracy rate. However, every time the actual class is background, the model is mistakenly predicting it as a seedling, leading to a high false positive rate. This means the model isn't good at differentiating between the seedling and background when the actual class is background.

Here some predicted images from YOLOv8 are shown.

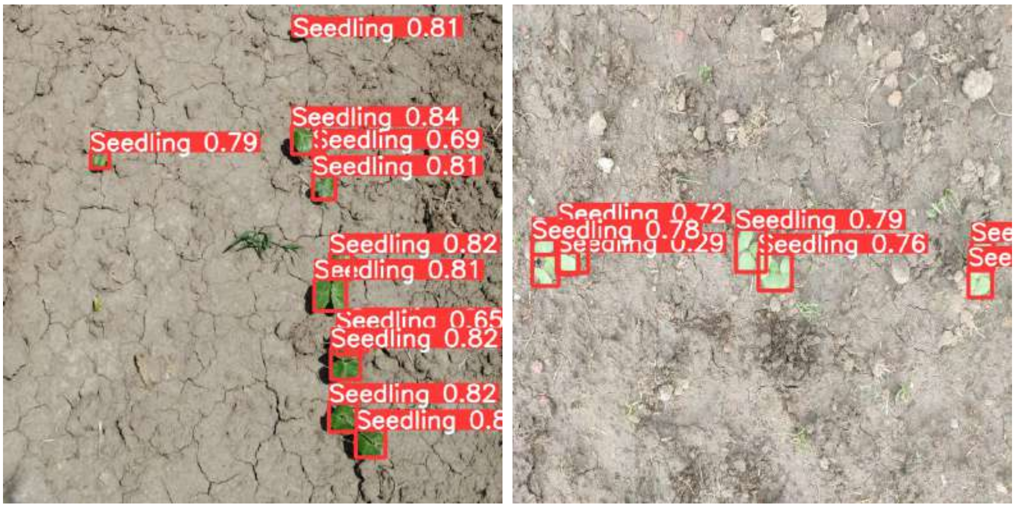


Figure 4.15: Predicted images with confidence score [YOLOv8]

Figure 4.16: Predicted images with confidence score [YOLOv8]

From the prediction, it is obvious that, our model effectively detected seedling with satisfactory confidence score.

4.4 Seedling segmentation on Cotton Seedling Dataset

The following table Table 4.5 shows how the models have performed on the test dataset.

Table 4.5: Model Performance Metrics

No.	Model	Batch Size	Epochs	Iterations	Best Epoch	bbox_mAP	segm_mAP
1	cascade-mask-rcnn_r50_fpn	2	12	10284	11	0.665	0.645
2	cascade-mask-rcnn_r50_fpn	1	20	34260	20	0.676	0.666
3	modified	1	20	34260	13	0.679	0.657
4	modified	2	10	8570	10	0.682	0.653

The model "cascade-mask-rcnn_r50_fpn" has been trained with both a batch size of 2 and 1, for 12 epochs and 20 epochs respectively. The first model achieved best bbox_mAP at epoch number 11 and second model at epoch number 20. The bounding box mean Average Precision (bbox_mAP) for model 1 was 0.665 and a segmentation mean Average Precision (segm_mAP) was 0.645. Model 2 improved because of some extra epochs training.

The modified model also has been trained with a batch size of 2 and 1. The third model was trained upto 20 epochs and achieved its best bbox_mAP at epoch number 13. The epoch number where model 4 shows best bbox_mAP was 10. The third model achieved a higher bounding box mean Average Precision (bbox_mAP) of 0.682 and a segmentation mean Average Precision (segm_mAP) of 0.653. But if we think of both the box mAP and the seg mAP as the criterion for judging the results, then the third model did the best. However, if the fourth model was given several more epoch trains, then this model has the potential to do better on both parameters. We agree that, here we have a limitation to proper comparison.

We could not give the fourth model more epoch train because we had to do this training process on Google Colab server. Due to memory shortage, it was not possible to train on our GPU.

Both the Model 3 and Model 4 were able to generate good results in low numbers of epochs suggesting better efficiency or faster convergence. That means the modifications made to the architecture or training process have resulted in a more effective and efficient model for the given task.

Here a question may arise as to why we reduced the batch size to 1 even though it was small. As we mentioned in methodology part, we have used SABLHead as bbox_head type in the new model which consumes a lot of memory. So we saw that if the batch size is to be done 1 then there is no issue on training. We had to reduce the batch size, though our purpose was not. Below is the graph of mAP and loss during the training phase of model 3.

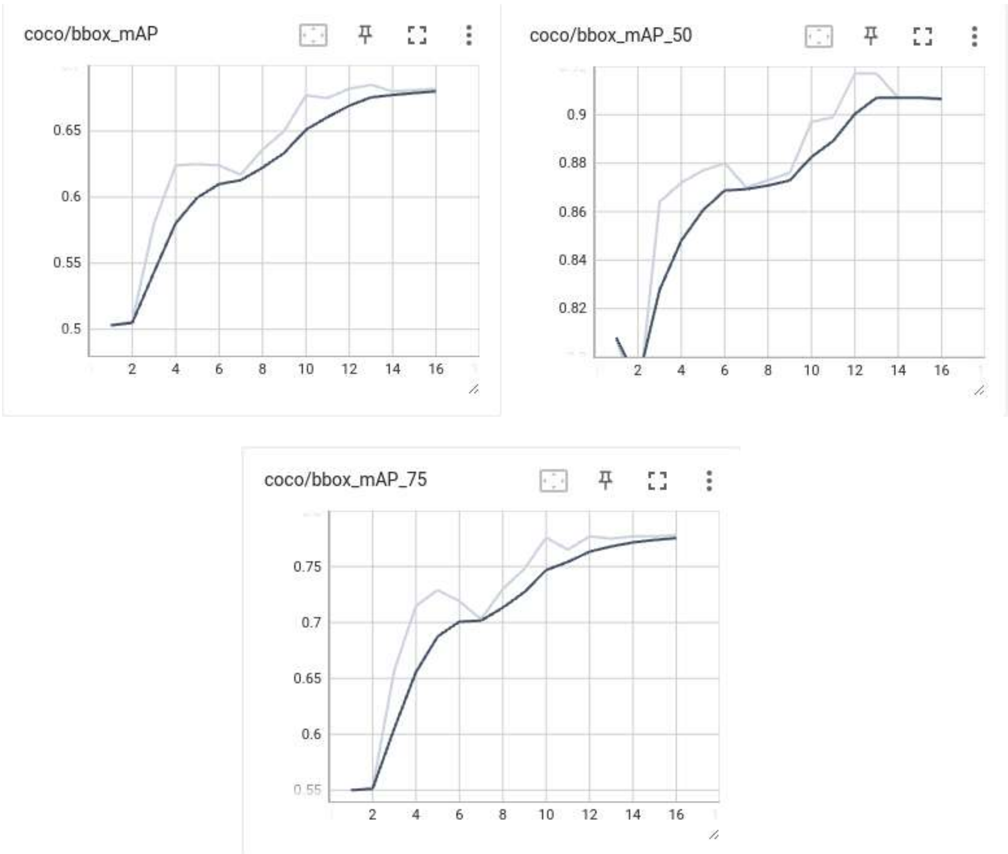


Figure 4.17: Average bbox_mAP (upper left), bbox_mAP at IoU=0.50 (upper right) and bbox_mAP IoU=0.75 (lower middle)

From the mAP graph, we can see that the mAP value was increasing from the beginning. However, towards the end, the growth slope was moving towards 0.

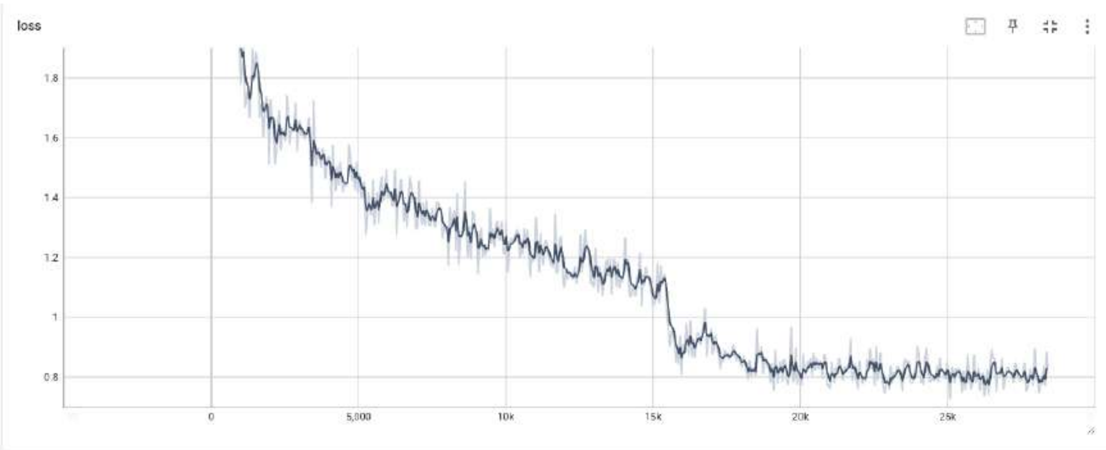


Figure 4.18: Training loss over 34k iterations

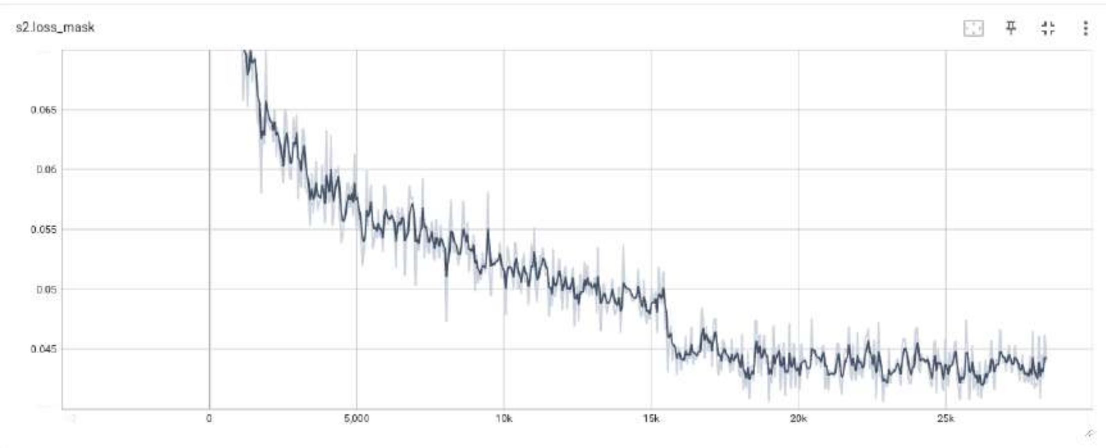


Figure 4.19: Training loss (mask) over 34k iterations

Here above is the box loss and mask loss of Model 3. Out of our four models training, model 4 shows the best box mAP, and in the case of seg mAP, model 2 shows the best result.

We think it is necessary to show the loss and box mAP of model 4 here. But the model 4 had to be trained in Google Colab for memory shortage. Due to the elapsed training time of more than 7 and a half hours, the training was stopped from the server. It was not possible to save the training data. If there was no memory shortage, we would have tried more epochs and it was not tough to draw a loss graph for model 4.

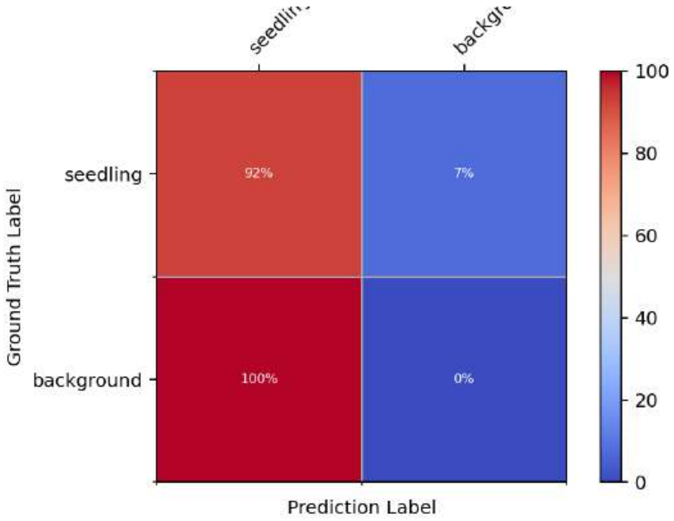


Figure 4.20: Confusion matrix

From the confusion matrix, we can observe that True Positive (TP) = 92%, False Positive (FP) = 0%, True Negative (TN) = 100%, False Negative (FN) = 7%. For the "seedling" class, the model correctly predicted 92% of instances as "seedling." For the "background" class, the model correctly predicted 100% of instances as "background" when the ground truth was also "background."

Figure 4.21: Test Image (left) and Output Image (right)

Figure 4.22: Test Image (left) and Output Image (right)

Figure 4.23: Test Image (left) and Output Image (right)

Chapter 5

Conclusion

In our project, we conducted an exploration of seedling detection and segmentation using three YOLO models (YOLOv5, YOLOv8, YOLONAS) and two versions of the Cascade R-CNN model (default and modified). Our study was grounded in a dataset we acquired from crop field, and the results provide valuable insights into the capabilities and applications of these state-of-the-art models in agricultural computer vision.

The YOLO models demonstrated efficiency in real-time seedling detection, with varying trade-offs between speed and accuracy. YOLOv5, YOLOv8, and YOLONAS showed their distinct strengths.

Our study examined also the part of seedling segmentation, employing both default and modified versions of the Cascade R-CNN model. Notably, the modified Cascade R-CNN outperformed its default counterpart, highlighting the significant impact of tailored optimizations on segmentation accuracy.

The uses of seedling detection and segmentation extend far beyond the realm of academic research. These technologies hold the potential to precision agriculture by providing farmers with actionable insights for informed decision-making. Accurate seedling detection aids in optimizing resource allocation, guiding interventions such as irrigation and fertilization, while precise segmentation enables a deeper understanding of plant structures and growth patterns.

Our findings underscore the importance of customization and fine-tuning in model development. The success of the modified Cascade R-CNN in seedling segmentation serves as a testament to the benefits of model optimization tailored to specific agricultural contexts.

The insights gained from our work provide valuable guidance for researchers and practitioners seeking to implement advanced computer vision techniques in precision agriculture. As we propel into the future, the knowledge gained here sets the stage for continued advancements, fostering sustainable and technologically-driven farming practices.

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